

Primes, Folds, and the One Dot

A visual atlas of the prime relay, the sieve geometry, the magenta walk, the zero horn, the light cone and the prime redshift

Sep 28, 2026 · @Christophe Michaels

This paper gathers every figure and computation from the working sessions of 27 and 28 September 2026 into one record. It does not prove the Riemann Hypothesis. It maps the object from many angles, checks every identity it uses to stated precision, and places the Weil energy program on the current proof frontier, where a new unconditional result (more than two thirds of zeros on the line) was built from the same machinery.

How to read this paper

Every statement carries one of four tags, following the Dispatch Method's claim discipline. Nothing here is promoted past its tag.

Tag	Meaning
Proven	A theorem in the published literature, cited in Sources.
Computed	A number produced in these sessions, with its precision and cross-check stated.
Conjecture	Widely believed and supported by data, not proved.
Interpretation	Christophe's framing language (octaves, colors, rules, the fourth dimension). It guides the pictures; it is not a mathematical claim.

The object under study is the completed zeta function $F(z) = \xi(1/2 + z)$ and its zeros. The Riemann Hypothesis (RH) says every zero has real part exactly $1/2$. RH stays **open** throughout. The paper's contribution is a unified set of pictures, a set of exact identities checked on the real function, several new measurements, and a map from the program's own objects (the odd Weil form Q_a , the infinity plane, the quadratic plane, the color relay) to the current proof frontier.

Figures are numbered 1 to 31 in reading order. Parts I to IV come from the first working session (27 September), Parts V to XIV from the second (27 to 28 September).

Part I. The prime relay and the color energies

Each prime power holds the odd Weil energy floor up exactly where it enters. Remove the newest prime power and the floor turns negative within a few thousandths of its entry point. **Computed**, high precision, not certified.

Setup. The odd Weil form Q_a on test functions supported in $[-a, a]$ has the Fourier symbol

$$\Psi_L(t) = \operatorname{Re} \left(\psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi - \sum_{\log n < 2L} \frac{\Lambda(n)}{\sqrt{n}} \cos(t \log n) \right)$$

plus the pole term $-2(\int f(x) \sinh(x/2) dx)^2$. A prime power n enters the form when $2a$ passes $\log n$. RH is equivalent to Q_a staying positive for every a (Weil's criterion). The engines were validated against the published reference floors of Zhu (support certified to 0.8): 1.5599×10^{-14} at $a = 0.8$ against Zhu's 1.57×10^{-14} , and 1.4913×10^{-26} at $a = 1.0$ against 1.49×10^{-26} .

Each prime holds the odd Weil floor up where it enters

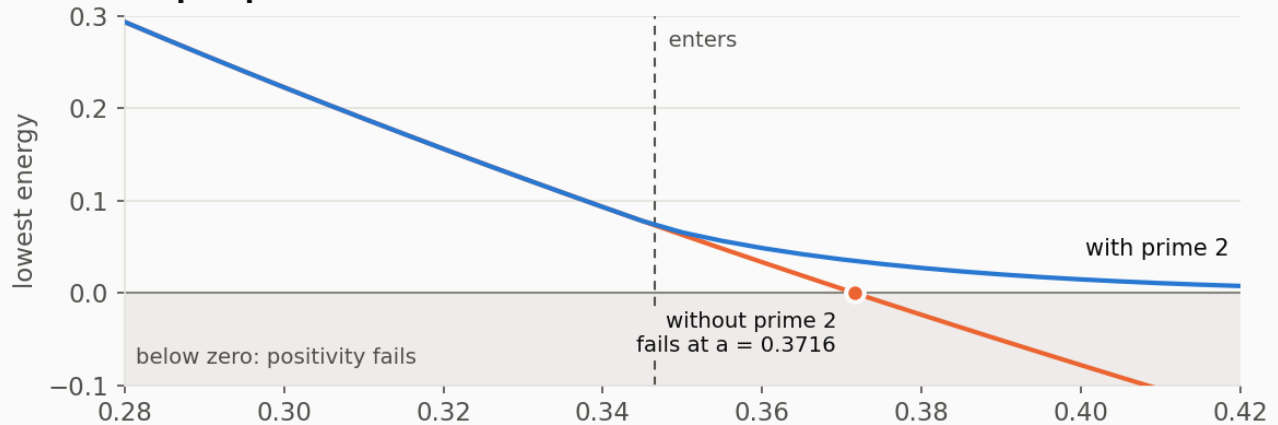
Lowest energy of the odd Weil form on $[-a, a]$, with and without the newest prime power.

Without it, the floor crosses zero just after the entry point.

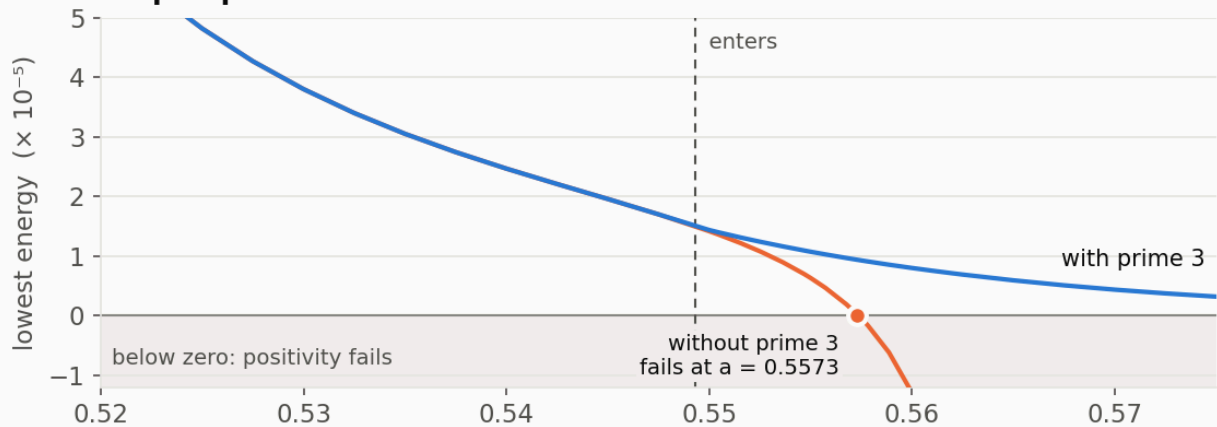
Double-precision numerics with 24 odd Legendre modes, not certificates.

— with the newest prime (the actual Weil form) — without it

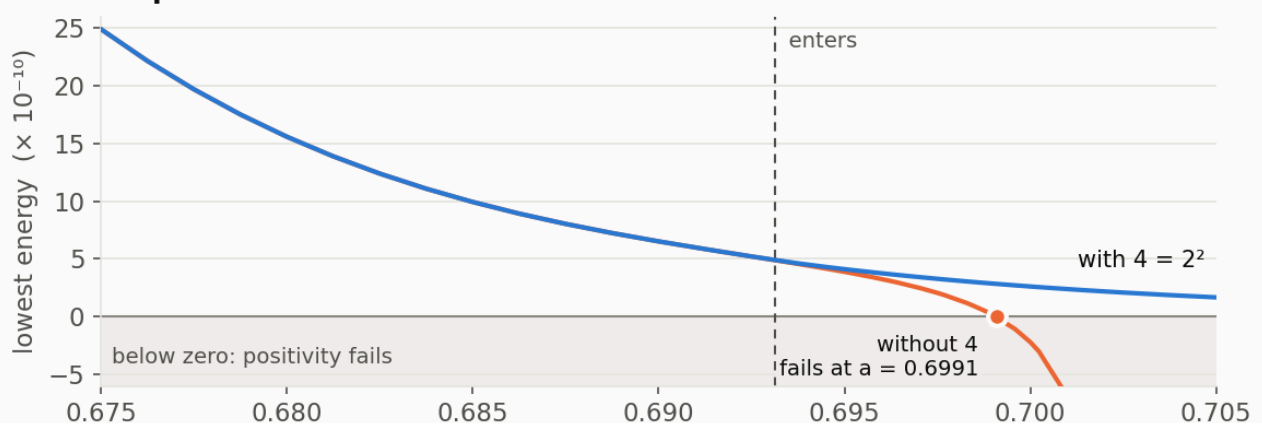
Step 1: prime 2 enters at $a = 0.347$



Step 2: prime 3 enters at $a = 0.549$



Step 3: $4 = 2^2$ enters at $a = 0.693$

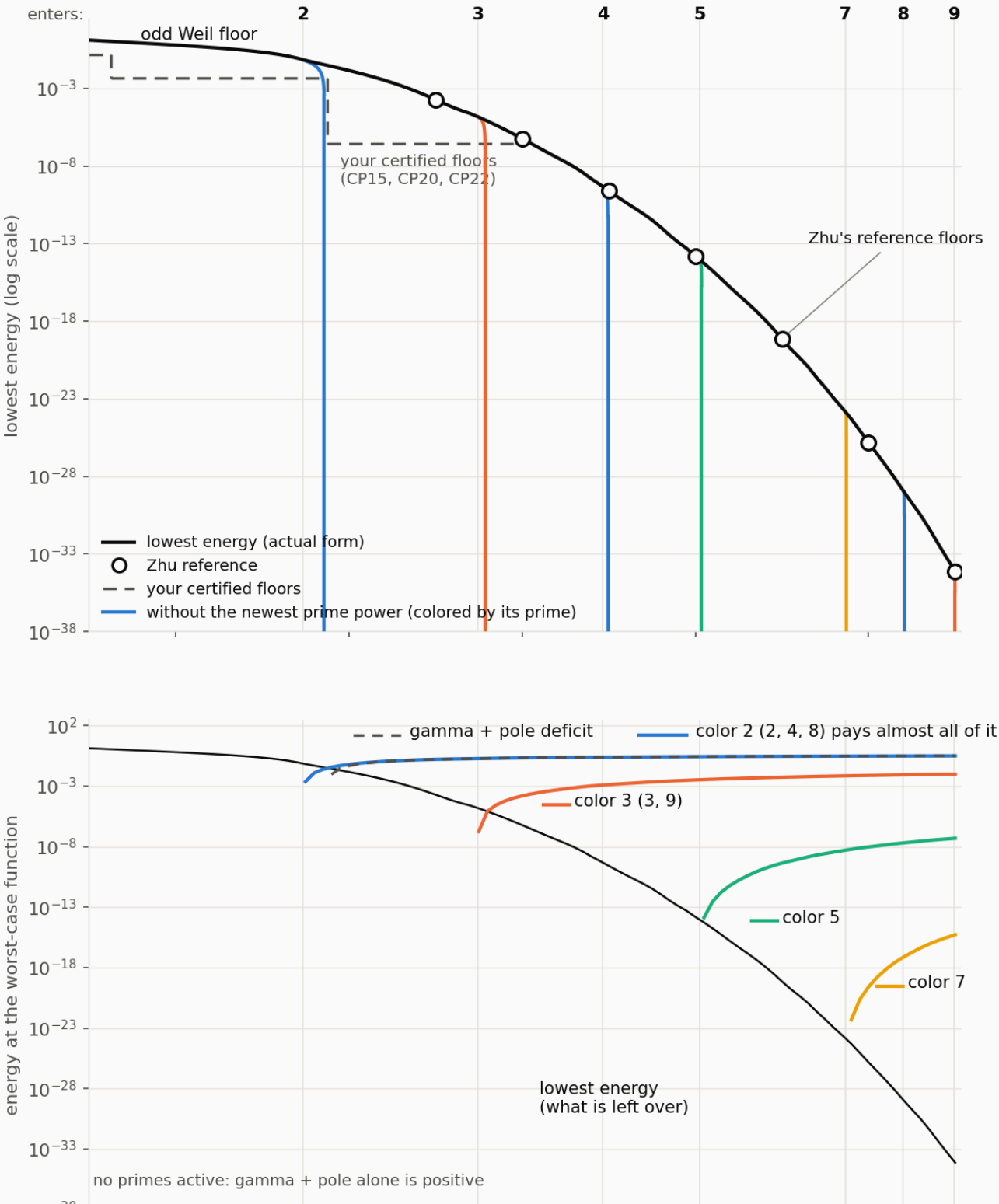


support half-width a

Figure 1. The relay, first three hand-offs. Lowest energy of the odd Weil form with the newest prime power (blue) and without it (orange). Without 2 the floor fails at $a = 0.3716$, without 3 at 0.5573, without $4 = 2^2$ at 0.6991.

Seven prime powers, seven hand-offs

Odd Weil form on $[-a, a]$, out to $a = 1.1$. Top: its lowest energy. Each colored drop is the same form without its newest prime power, which fails just after that prime power enters.
Bottom: at the worst-case function, the gamma factor plus pole run a deficit and the prime colors (all powers of p grouped, as in H7) pay it. High-precision numerics, not certificates.



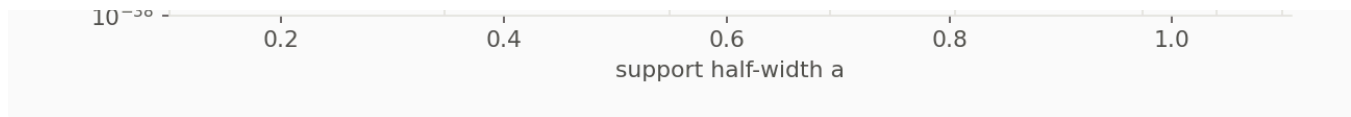


Figure 2. Seven prime powers, seven hand-offs, out to $a = 1.1$. Top: the floor on a log scale with Zhu's reference points and the program's certified floors (CP15, CP20, CP22). Bottom: at the worst-case test function the gamma factor plus pole run a deficit, and the prime colors pay it.

Prime power	Enters at $a = (\log n)/2$	Floor fails without it at	Gap	Precision
2	0.346574	0.371601	0.0250	420 digits, 40 modes
3	0.549306	0.557323	0.0080	420 digits, 40 modes
$4 = 2^2$	0.693147	0.699032	0.0059	420 digits, 40 modes
5	0.804719	0.807245	0.0025	640 digits, 72 modes
7	0.972955	0.974246	0.0013	640 digits, 72 modes
$8 = 2^3$	1.039721	1.041389	0.0017	640 digits, 72 modes
$9 = 3^2$	1.098612	1.099731	0.0011	640 digits, 72 modes

The color budget. Group every power of p into one color. At the worst-case function, color 2 (2, 4, 8) pays 98.8% of the gamma plus pole deficit at $a = 0.8$ and 97.6% at $a = 1.0$. Color 3 pays nearly all the rest (2.4% at $a = 1.0$); colors 5 and 7 contribute 3×10^{-8} and 9×10^{-20} of the deficit at $a = 1.0$. **Computed.**

What the relay is and is not. It is a receipt: at each tested support, the newest prime power is load-bearing. It is not yet a rule: nothing here proves the hand-off continues for every a , and the floor itself falls super-exponentially (Part XII). **Interpretation:** each prime is a strut placed just in time.

Part II. Music of the primes: the zeta tuning and the descent law

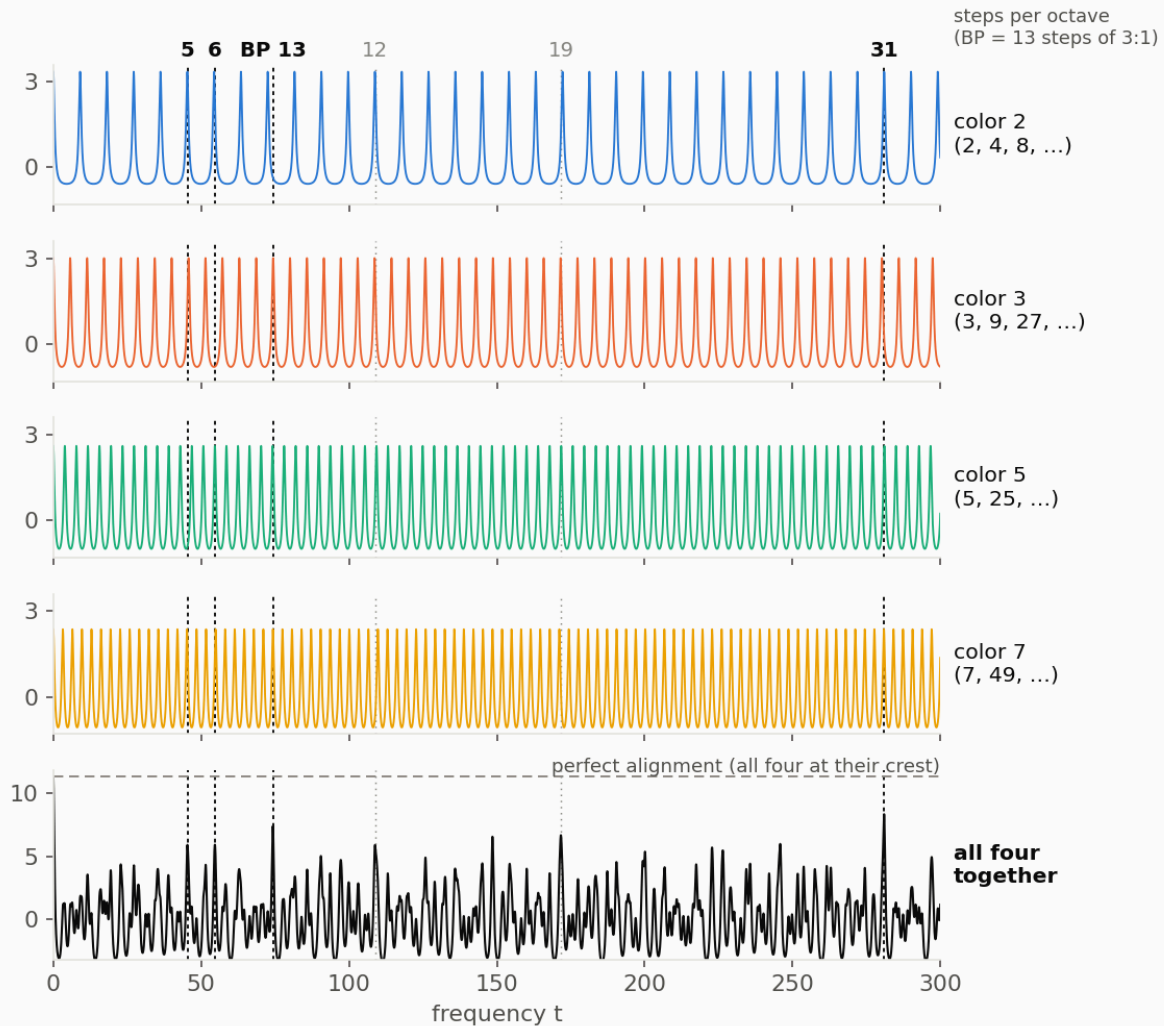
The Weil comb is a chord of prime colors, and the frequencies where the colors crest together are equal temperaments of the octave. How closely they can ever align follows Dirichlet's approximation law, the same law random numbers obey. **Computed**; the law itself is **Proven** (Dirichlet).

The tuning dictionary. At frequency $t = 2\pi k / \log 2$, color 2 (every power of 2) sits exactly at its crest: k steps per octave. The other colors crest there only approximately; the miss is a comma. Where all colors crest together, the prime comb peaks and the Weil symbol dips.

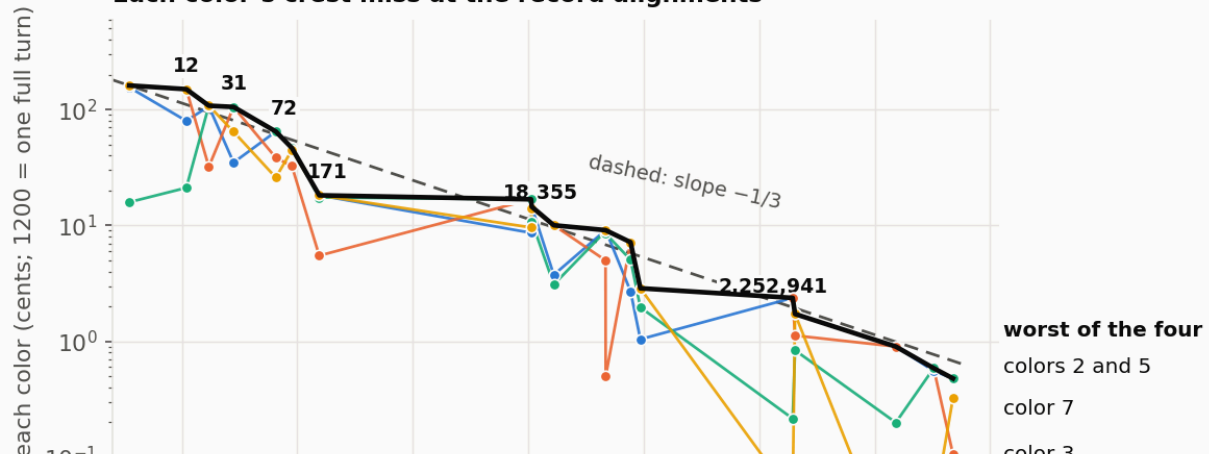
The 2, 3, 5 and 7 lines

Top: each color's wave in the Weil comb (your H7 ladder, all powers of p together), against frequency t . Where they crest together the comb peaks and the Weil symbol dips. Dark dotted lines: new record highs of the four together. Light: 12 and 19 steps per octave, for reference.

Bottom: at every record alignment up to $t \approx 5 \times 10^8$, how far each color misses its crest.



Each color's crest miss at the record alignments



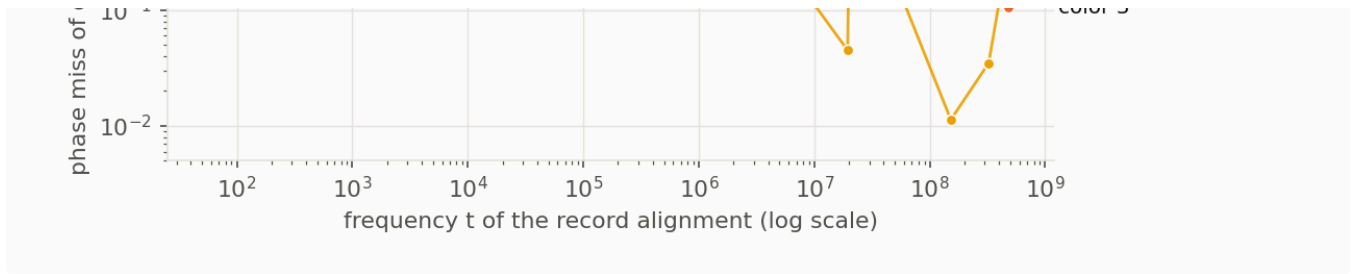
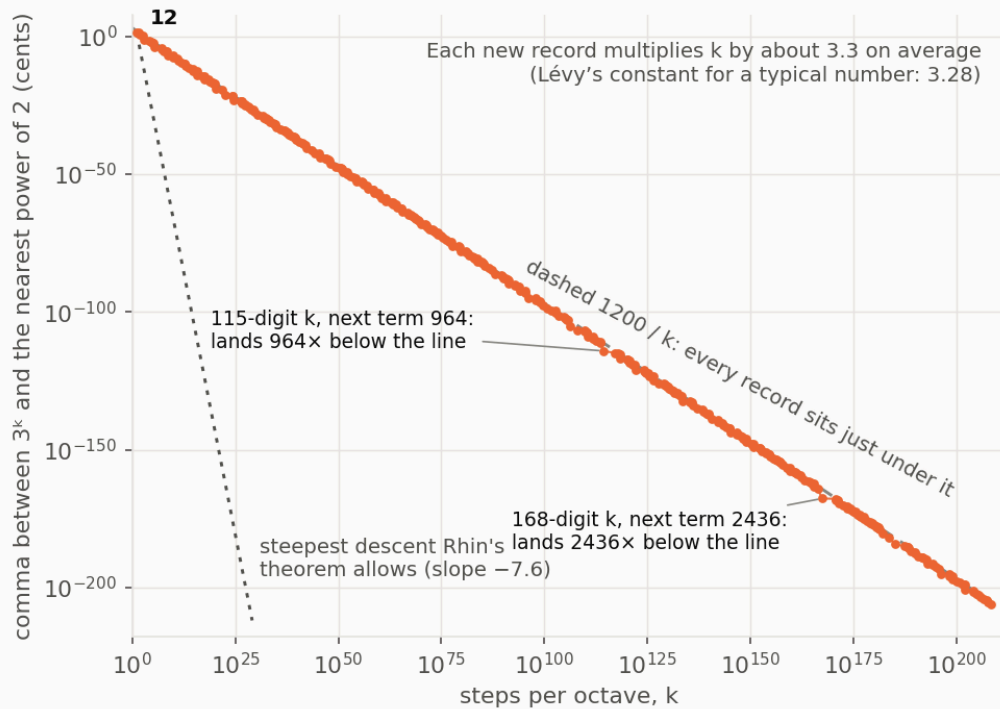


Figure 3. The 2, 3, 5 and 7 lines. Top: each color's wave in the Weil comb against frequency t , with record alignments of all four marked (5, 6, the Bohlen–Pierce 13 steps of 3:1 near $t = 74.26$, and 31). Bottom: at every record alignment up to $t \approx 5 \times 10^8$, how far each color misses its crest; the worst of the four follows slope $-1/3$.

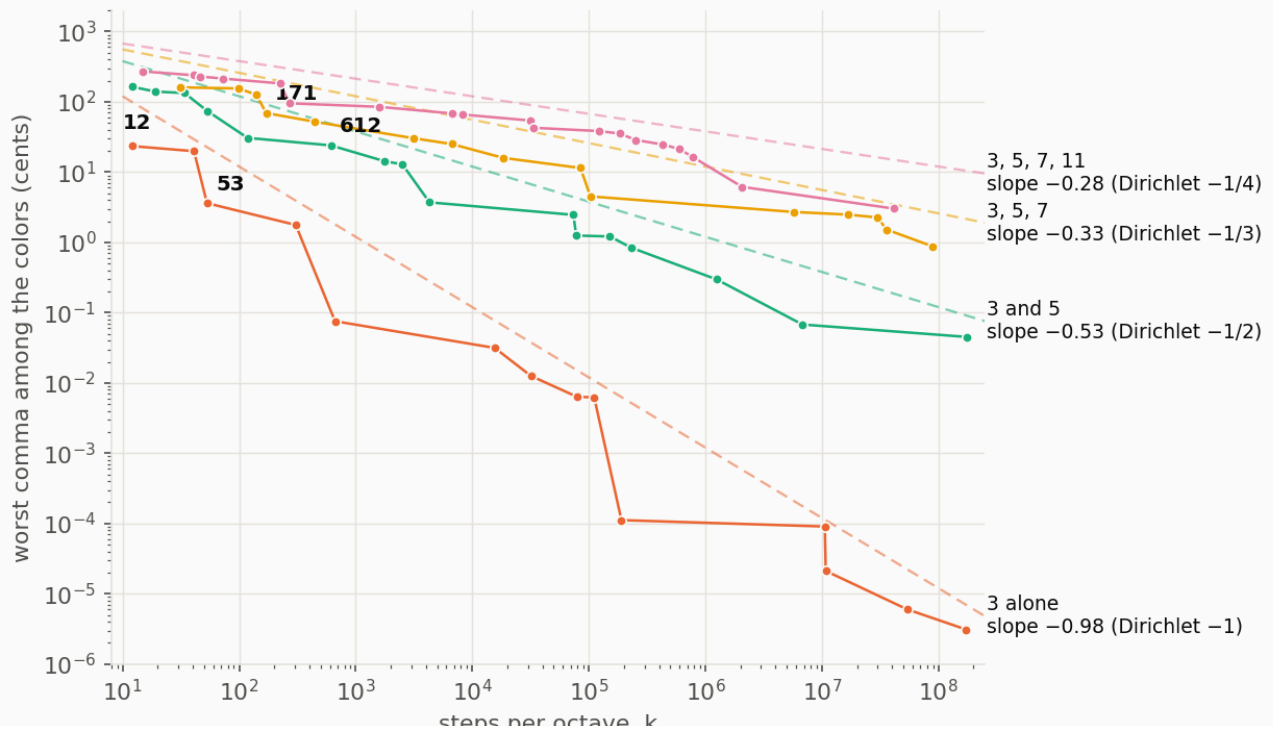
How fast the chord closes

Each point is a record tuning: an equal division of the octave that brings the colors closer to alignment than any smaller one. The comma is the phase miss each color leaves in the Weil comb at that tuning's frequency, $t = 2\pi k / \log 2$.

Octave and 3 alone: 399 record tunings, out to 209-digit step counts



More colors: every tuning up to 200 million steps



Bottom panel, dashed lines: Dirichlet's guarantee, $1200 \cdot k^{(-1/d)}$ for d colors.
 Bold numbers: tunings where the Weil symbol dipped in the earlier scan.

Figure 4. *How fast the chord closes. Top: the octave and 3 alone, 399 record tunings out to 209-digit step counts; every record sits just under $1200/k$ cents, each new record multiplies k by about 3.3 on average (Lévy's constant for a typical number is 3.28), far from the steepest descent Rhin's theorem allows. Bottom: more colors, every tuning up to 200 million steps.*

Colors aligned with the octave	Record tunings (steps per octave)	Records to 2×10^8	Fitted slope	Dirichlet exponent
3	5, 12, 41, 53, 306, 665, 15601, 31867, ...	17	-0.98	-1
3, 5	10, 12, 19, 34, 53, 118, 612, 1783, ...	20	-0.53	-1/2
3, 5, 7	9, 10, 31, 99, 140, 171, 441, 3125, ...	20	-0.33	-1/3
3, 5, 7, 11	9, 15, 41, 46, 72, 224, 270, 1578, ...	23	-0.28	-1/4

Reading. The worst comma among d colors shrinks like k to the power $-1/d$. A test with random numbers in place of $\log 3$, $\log 5$, $\log 7$ gives the same slopes, so the descent is generic, not special to primes. What is special is that the colors can never align exactly: the logarithms of distinct primes are linearly independent over the rationals (unique factorization), so the comb never collapses onto a single crest. **Interpretation:** the primes form a chord that never quite resolves, and the Weil symbol's dips are its near-resolutions.

Part III. Two sides of the coin

The primes and the zeros are one object seen from two sides. Put every prime power on one line at position $\log n$ and let it ring; subtract all of them from the factorial's energy; what is left is a row of bumps sitting exactly at the zeta zeros. Run it backward and the zeros alone rebuild every prime bar. **Proven** (the Weil explicit formula); both directions **Computed**.

Two sides of the same coin

Top: every prime power placed on one line at $\log n$, each ringing as a wave (height $2\Lambda(n)/\sqrt{n}$).
 Bottom: the factorial's energy minus all 78,734 prime-power waves up to one million (solid), against Gaussian bumps placed at the zeta zeros alone (dashed). Numbers mark the zero heights.

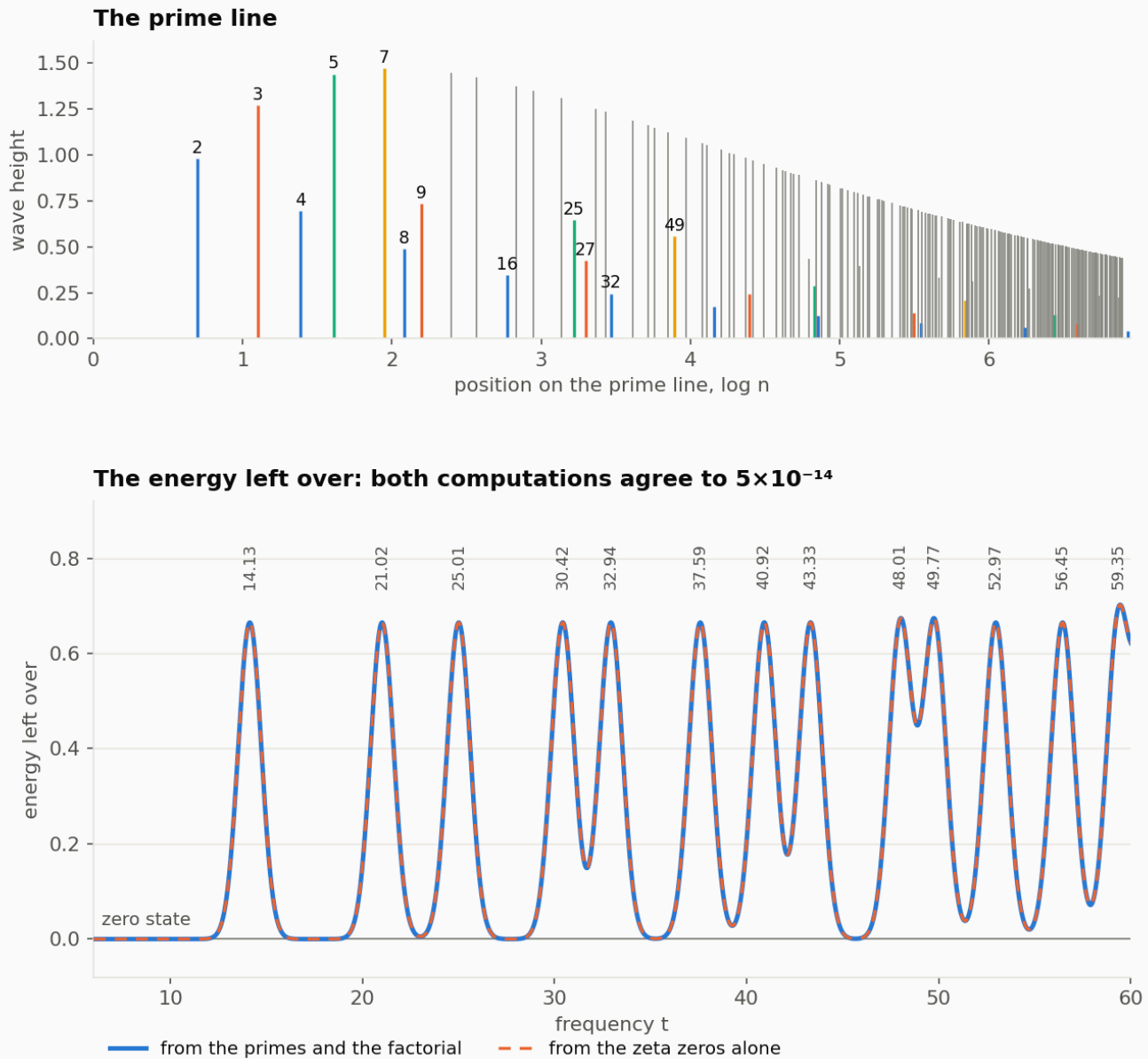


Figure 5. Two sides of the same coin. Top: the prime line, every prime power at $\log n$ with wave height $2\Lambda(n)/\sqrt{n}$. Bottom: the factorial's energy minus all 78,734 prime-power waves up to one million (solid) against Gaussian bumps placed at the zeta zeros alone (dashed). The two agree to 5×10^{-14} .

The factorial enters through Chebyshev's identity, $\log n! = \sum \Lambda(d) [n/d]$, which is how the gamma factor in the Weil symbol carries the "counting numbers" side.

The envelope. Christophe drew a magenta envelope over the prime line by hand. It is exact: every prime sits on $y = 2x \cdot e^{(-x/2)}$ with $x = \log p$, every square on half of it, every cube on a third, every k -th power on $1/k$.

$$\frac{2\Lambda(p^k)}{\sqrt{p^k}} = \frac{1}{k} \cdot 2x \cdot e^{-x/2}, \quad \text{where } x = \log p^k$$

Your curve is exact

Every prime sits on $y = 2x \cdot e^{(-x/2)}$, with $x = \log p$. Every square sits on exactly half of it, every cube on a third, every fourth power on a quarter: the $1/k$ ladder of a sawtooth wave's overtones.

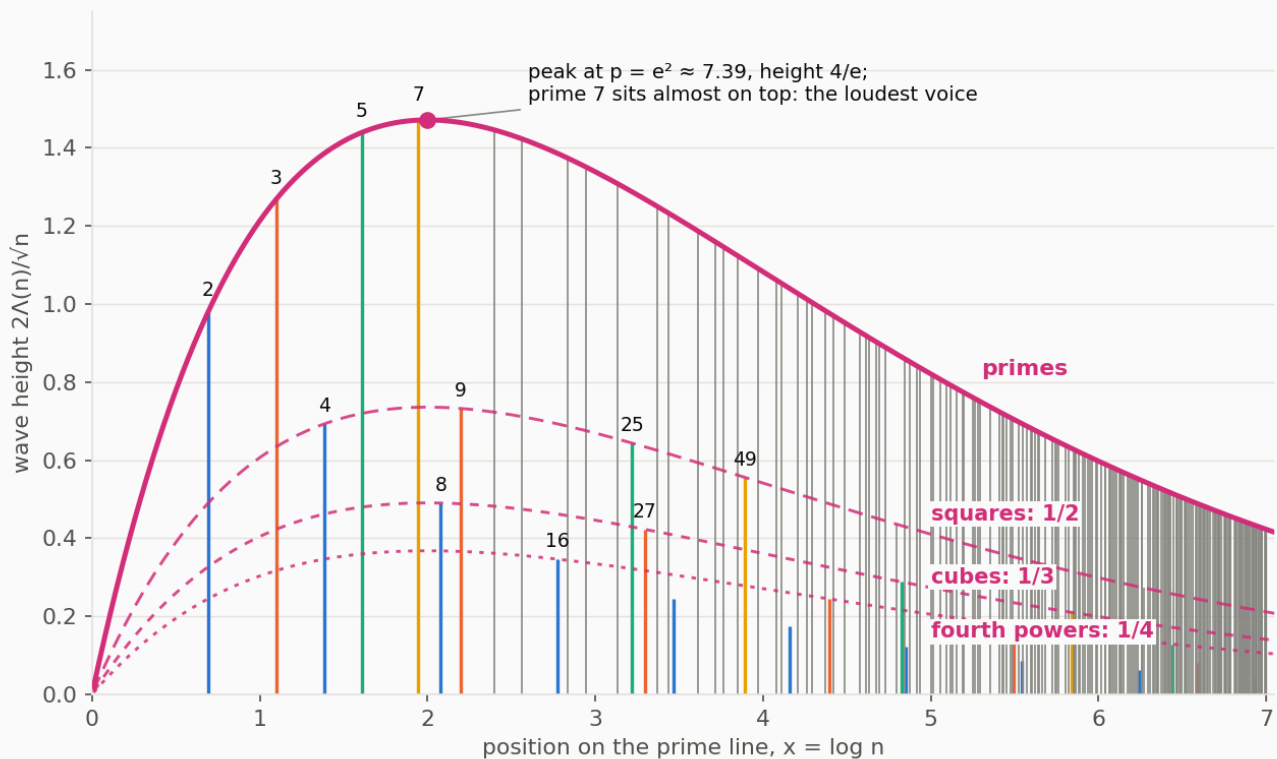
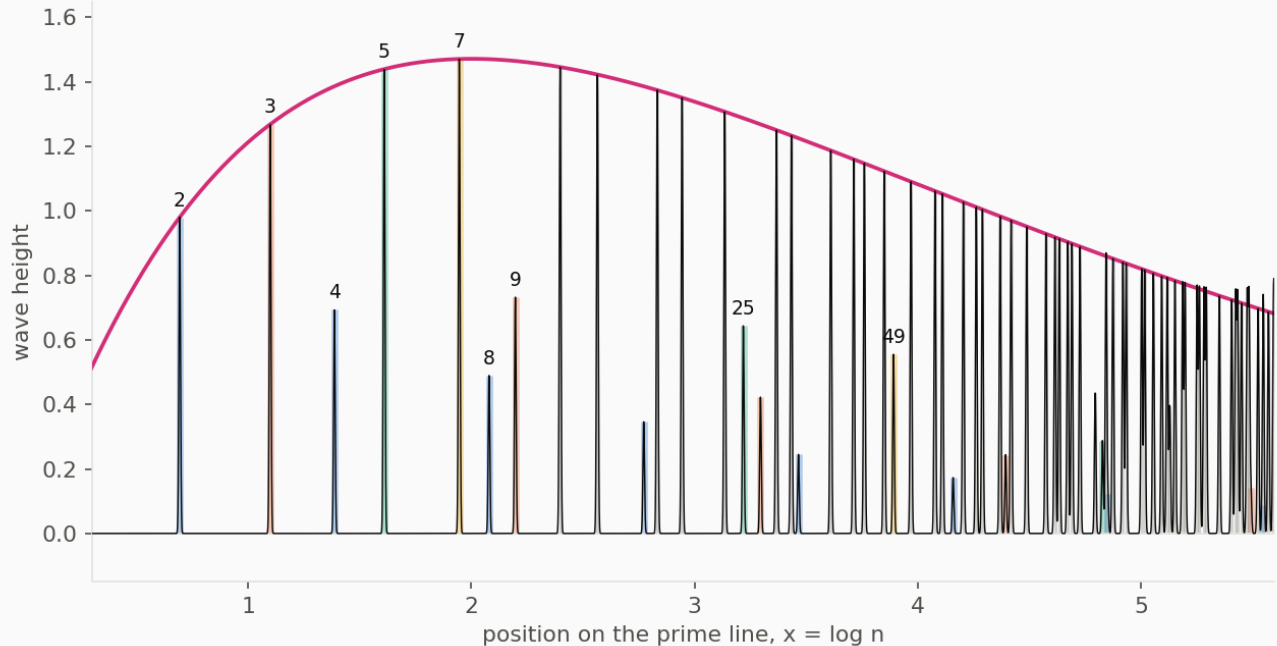


Figure 6. The envelope. Primes on $2x \cdot e^{(-x/2)}$, squares on $1/2$ of it, cubes on $1/3$, fourth powers on $1/4$: the overtone ladder of a sawtooth wave. The peak is at $p = e^2 \approx 7.39$ with height $4/e$; prime 7 sits almost on top, the loudest voice.

The zeros rebuild the prime line

Faint bars: every prime power at $\log n$, height $2\Lambda(n)/\sqrt{n}$. Magenta: your curve. Black: a line built only from the zeta zeros: the smooth choir $2e^{(x/2)}$, minus one cosine wave for each zero. It rises to the top of every bar and falls back to zero between them. No prime was used to draw it.

First 1,000 zeros, out to $n \approx 270$ (where they still separate neighboring primes)



Zoom: the picture sharpens as zeros are added

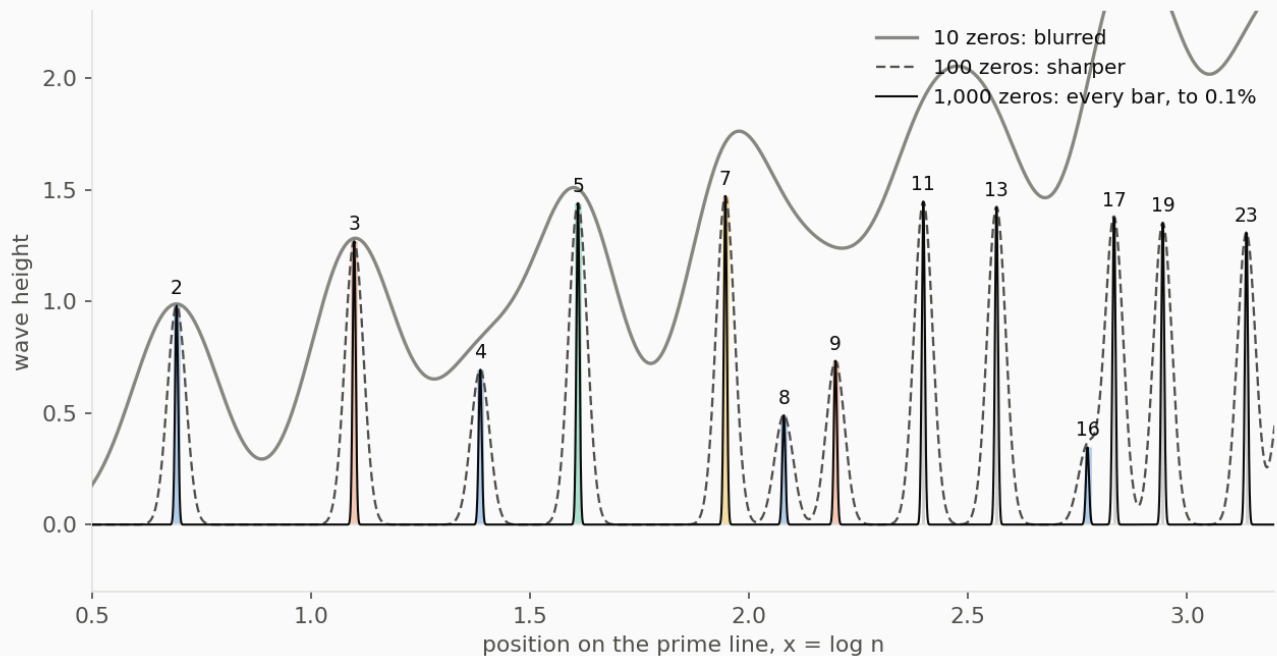


Figure 7. The zeros rebuild the prime line. Black: a line built only from the zeta zeros (the smooth term $2e^{(x/2)}$ minus one cosine wave per zero). No prime was used to draw it, yet it rises to the top of every prime-power bar and falls to zero between them. With 1,000 zeros every bar out to $n \approx 270$ is matched to 0.1% (1.6×10^{-6} in the smooth-weighted check); 10 zeros give a blur, 100 a sharper picture.

Interpretation: "the rules do not care about the value of a prime, only about its place in succession" is the explicit formula's own shape: the zero side sees each prime power only through its position $\log n$ and its weight, never through its digits.

Part IV. The parabola sieve in the color system

The Matiyasevich–Stechkin sieve turns multiplication into geometry: the chord from (a^2, a) to $(b^2, -b)$ on the parabola $x = y^2$ crosses the axis at exactly $a \cdot b$. Every composite is hit by some chord; the primes are the points no chord reaches. **Proven** (elementary algebra: the chord is $x = ab + (a - b)y$).

$$x = ab + (a-b)y \quad \Longleftrightarrow \quad x|_{y=0} = ab, \quad \text{angle to the axis} = \arctan \frac{1}{|a-b|}$$

The program's own note (AN01, AN02) adds the angle reading: chords with the same difference $a - b$ run parallel, so the angle records the gap between the two factors, not their values.

The sieve in your colors

Each chord joins (a^2, a) to $(b^2, -b)$ and crosses the axis at $a \cdot b$. It is drawn as a braid: one strand for each prime factor of $a \cdot b$, in that prime's color, so multiplication adds color vectors. Braids with the same difference $a - b$ run parallel (angle $\arctan 1/|a - b|$). Open circles on the axis are the primes: the positions where no braid lands. Braids from 8 through 24 enter from off the frame.

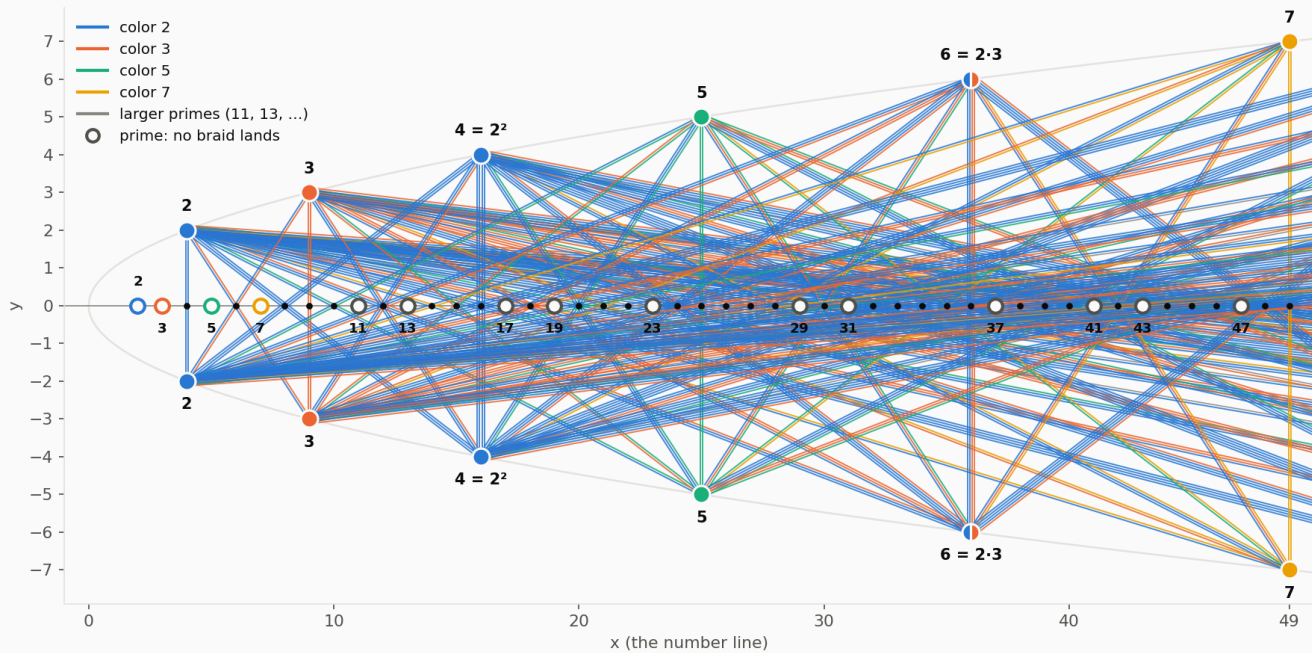


Figure 8. The sieve in the color system, drawn as braids: one strand per prime factor of $a \cdot b$ in that prime's color, so multiplication adds color vectors. Open circles on the axis are the primes.

The sieve, one color at a time

Each panel is one prime's rule: lines from (p^2, p) to every partner $(b^2, -b)$, crossing the axis at $p \cdot b$.

Top half of each line: the prime's color. Bottom half: the partner's smallest prime color (gray for 11 and up).

Dots mark the numbers that rule hits. Bottom panel: all four rules together. Open circles are the primes, hit by no rule.

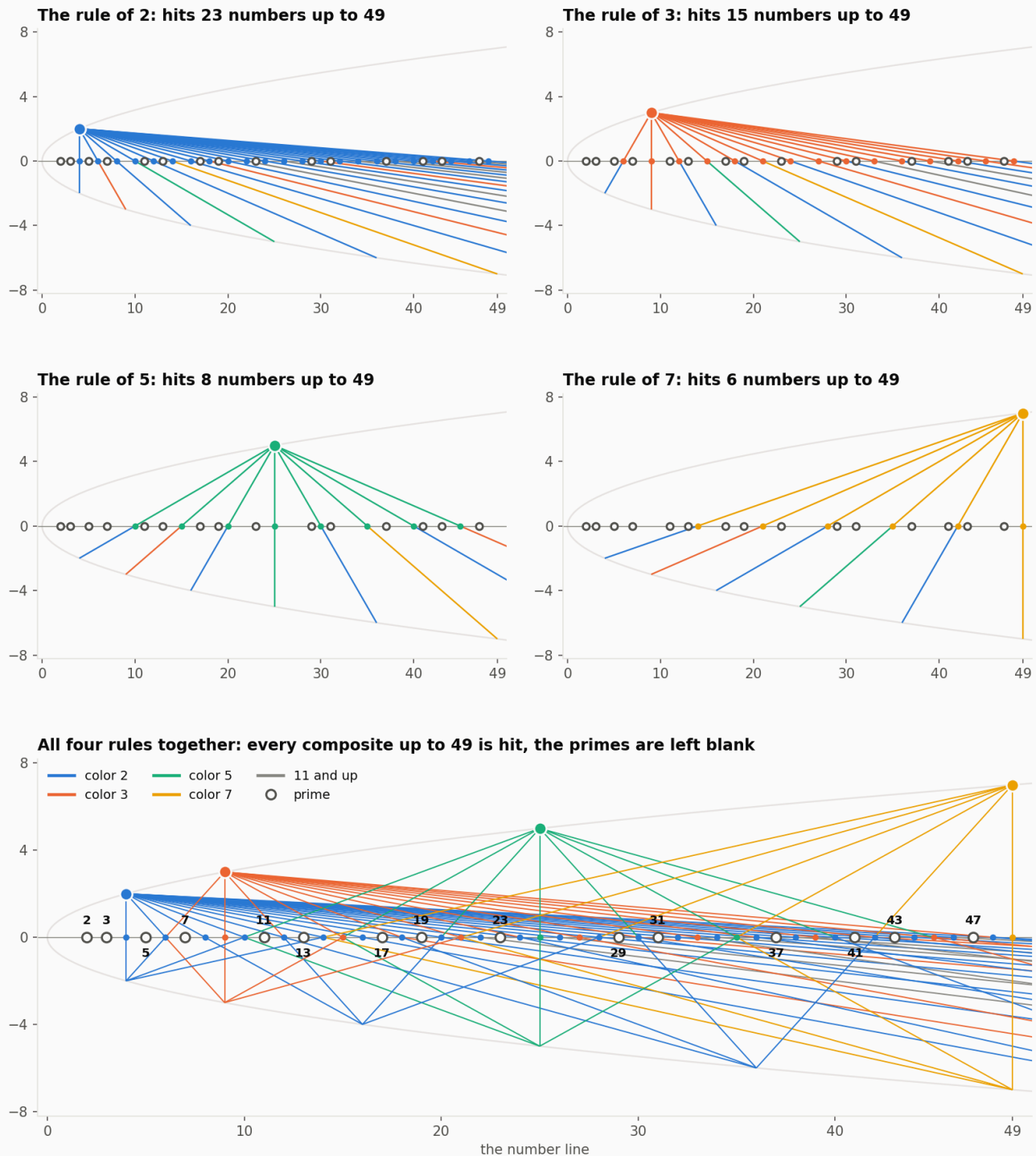


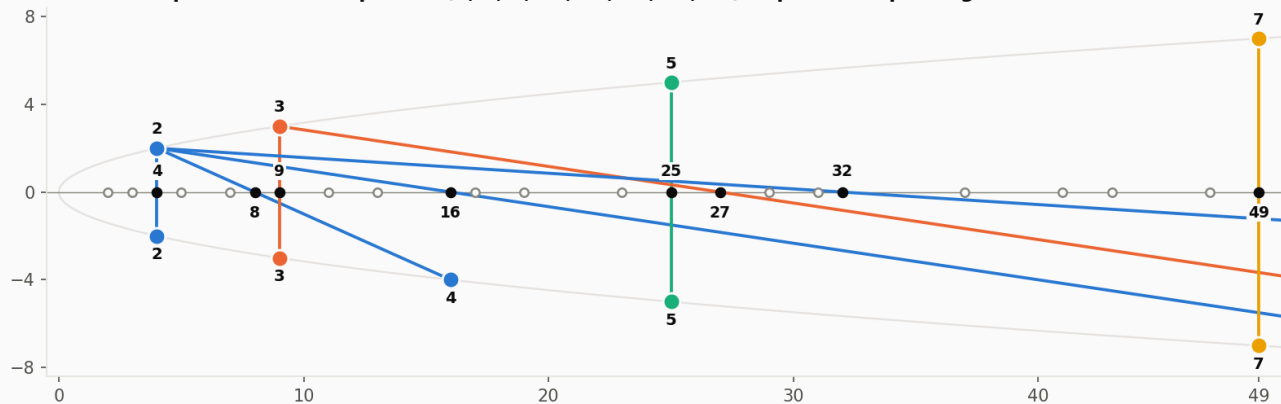
Figure 9. One rule per panel. Lines from (p^2, p) to every partner $(b^2, -b)$; top half in the prime's color, bottom half in the partner's smallest prime color. Up to 49 the rule of 2 hits 23 numbers, the rule of 3 hits 15, the rule of 5 hits 8, the rule of 7 hits 6. Together they hit every

composite and leave the primes blank.

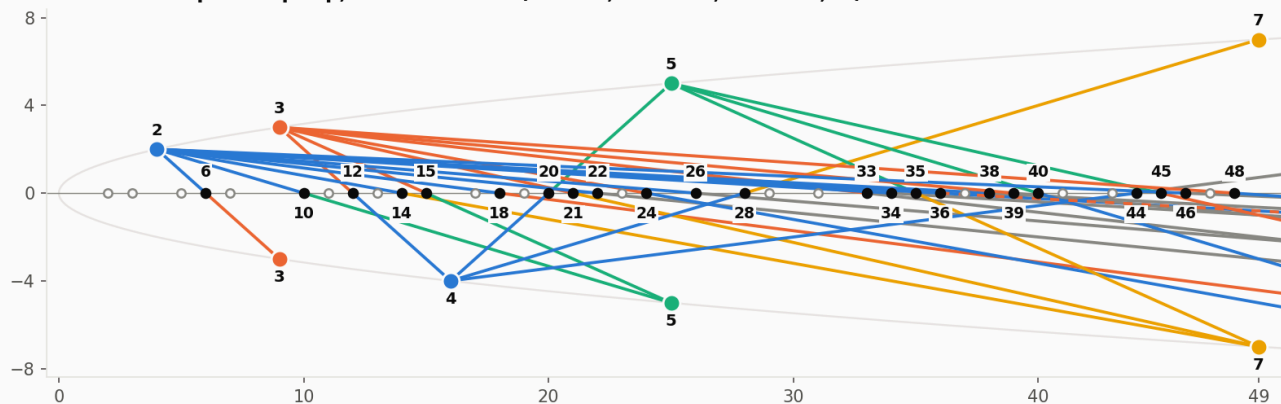
The sieve sorted by color

Only primes sit on top. Each line joins a prime (p^2 , p) to a bottom point (b^2 , $-b$) and lands on $p \cdot b$. Top half in the prime's color, bottom half in the color of b . A bottom point that carries two colors is drawn solid with its second color dashed over it. Gray open circles are the primes. Points past 7 sit off the frame, so their lines enter from the edge.

One color up and down: the powers (4, 8, 9, 16, 25, 27, 32, 49). Squares drop straight down.



Two colors: a prime up top, the rest below ($2 \cdot 3 = 6$, $3 \cdot 4 = 12$, $2 \cdot 9 = 18$, ...)



Three colors: all three primes reach one point ($30 = 2 \cdot 15 = 3 \cdot 10 = 5 \cdot 6$; $42 = 2 \cdot 21 = 3 \cdot 14 = 7 \cdot 6$)

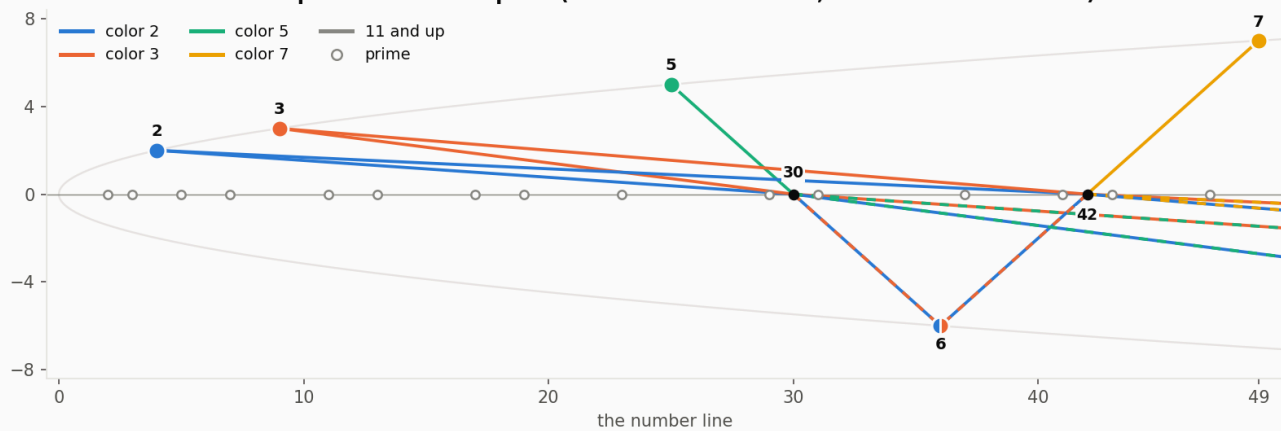


Figure 10. Sorted by type. Top: one color up and down, the prime powers (4, 8, 9, 16, 25, 27, 32, 49); squares drop straight down. Middle: two colors, a prime on top and the partner below (6, 12, 18, ...). Bottom: three colors meeting at one point (30 = 2·15 = 3·10 = 5·6; 42 = 2·21 = 3·14 = 7·6).

The full sieve: primes on top, in your colors

Only primes sit on top at (p^2, p); any counting number b sits below at ($b^2, -b$). Each line lands on $p \cdot b$, so every composite up to 49 is reached once from each of its primes. Top half of a line: the prime's color. Bottom half: the color of b , with a second color dashed over it when b carries two. Axis markers are filled with each number's own colors. Open gray circles are the primes: no line lands there.

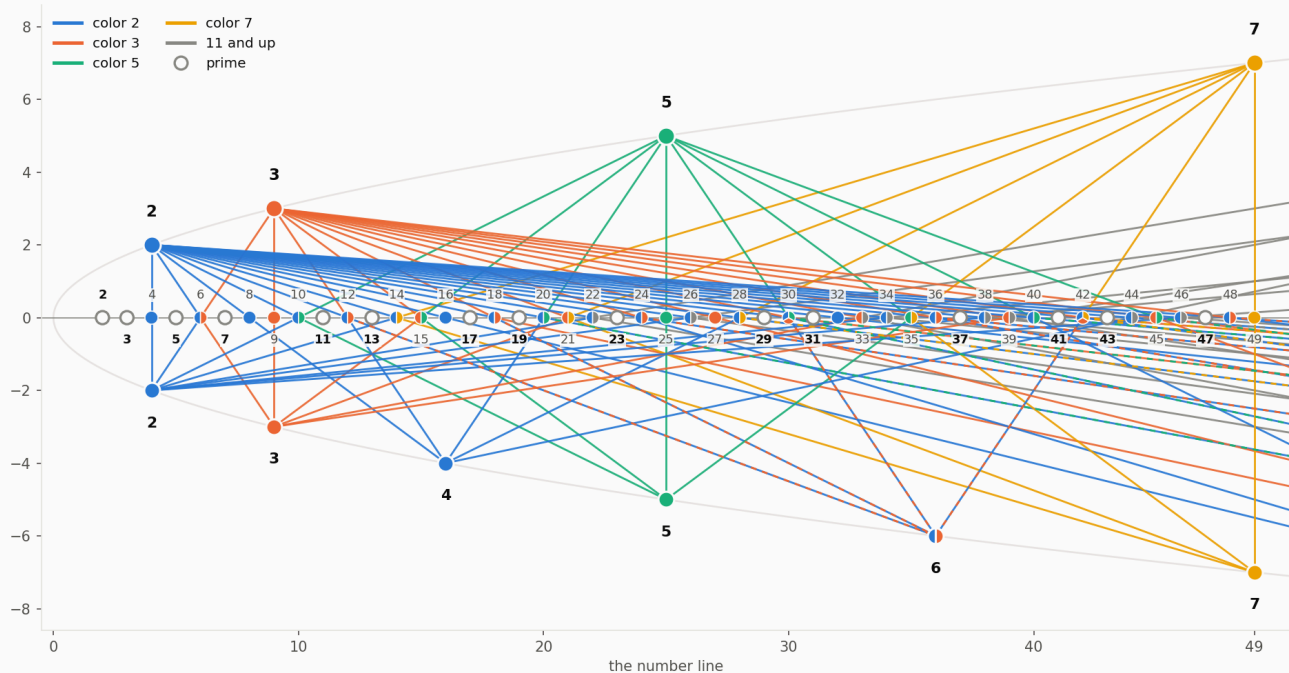


Figure 11. The full sieve with only primes on top: 60 lines, every composite up to 49 reached once from each of its prime factors. Top half of each line in the prime's color, bottom half in the partner's colors; axis markers are filled with each number's own colors, primes are open gray circles.

Counts. Each rule's first new composite is its square: new hits up to 49 are 23 for the rule of 2, 7 for the rule of 3, 2 for the rule of 5 and 1 for the rule of 7. **Computed;** the square threshold is **Proven** (every smaller multiple of p has a smaller prime factor).

Interpretation: in Christophe's words, each prime is an Uno card: the rule reads position and succession, not face value. The geometry agrees in one precise way: a chord's angle depends only on the difference of its two factors.

Part V. Natural cutoffs: the prime-square staircase

Christophe's hand-drawn magenta curve over the full sieve is the exact upper edge of its top half, and its natural cutoffs are the prime squares. Carried far enough out, the edge becomes an exact staircase whose step widths are the prime gaps. **Computed** (edge), **Proven** (staircase formula, elementary).

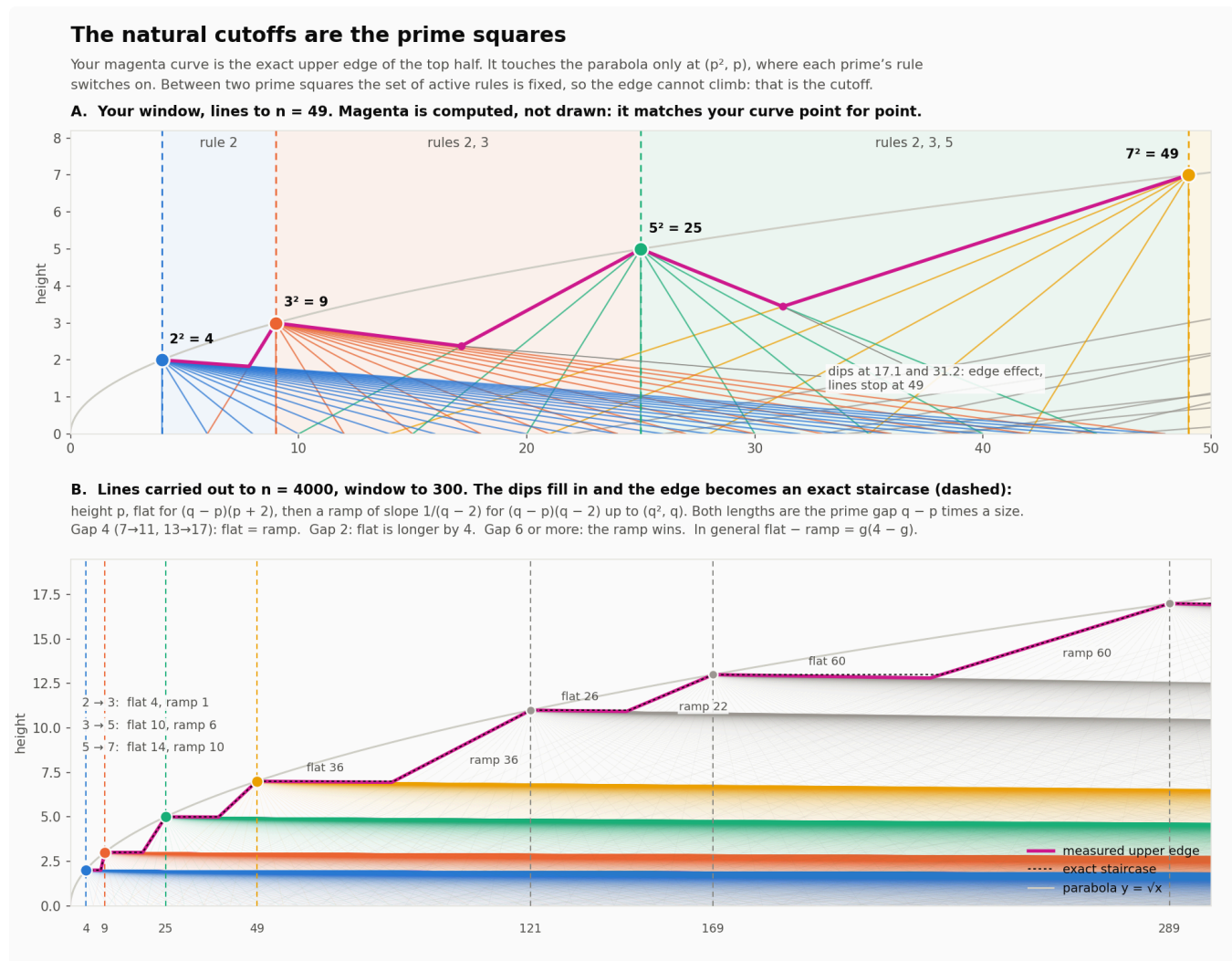


Figure 12. A: the computed upper edge for lines up to 49 matches the hand-drawn curve point for point: peaks at 4, 9, 25, 49 (where it touches the parabola), dips at 7.83 (height 1.83), 17.13 (2.38) and 31.22 (3.44), all edge effects of stopping at 49. B: with lines carried to 4,000 the dips fill in and the edge becomes an exact staircase.

Why the cutoffs sit at p^2 . A prime's rule adds its first new composite at p^2 . Between two consecutive prime squares the set of active rules cannot change, so the edge cannot climb. This is the sieve's classical $\sqrt{N}$ rule drawn by the geometry: to sieve up to N , primes up to $\sqrt{N}$ suffice.

The staircase. For consecutive primes $p < q$ with gap $g = q - p$, the edge sits flat at height p , then ramps to (q^2, q) along q 's shallowest chord (slope $1/(q - 2)$):

$$\text{flat} = (q-p)(p+2), \quad \text{ramp} = (q-p)(q-2), \quad \text{flat} - \text{ramp} = g(4-g)$$

$p \rightarrow q$	Gap	Flat	Ramp	Flat minus ramp
$2 \rightarrow 3$	1	[4, 8], 4	[8, 9], 1	3
$3 \rightarrow 5$	2	[9, 19], 10	[19, 25], 6	4
$5 \rightarrow 7$	2	[25, 39], 14	[39, 49], 10	4
$7 \rightarrow 11$	4	[49, 85], 36	[85, 121], 36	0
$11 \rightarrow 13$	2	[121, 147], 26	[147, 169], 22	4
$13 \rightarrow 17$	4	[169, 229], 60	[229, 289], 60	0

Gap 2 makes the flat longer by 4; gap 4 makes flat and ramp equal; gap 6 or more lets the ramp win. The measured edge with lines to 4,000 matches the formula to 0.072 over [4, 180].

Where the zeros enter. How long a flat can get is a prime-gap question. Under RH, Cramér's conditional bound gives gaps of order $\sqrt{p} \cdot \log p$, so no flat can much exceed $p^{3/2} \cdot \log p$. Unconditionally the best known gap exponent is 0.525 (Baker, Harman and Pintz). **Proven** (both, in the literature).

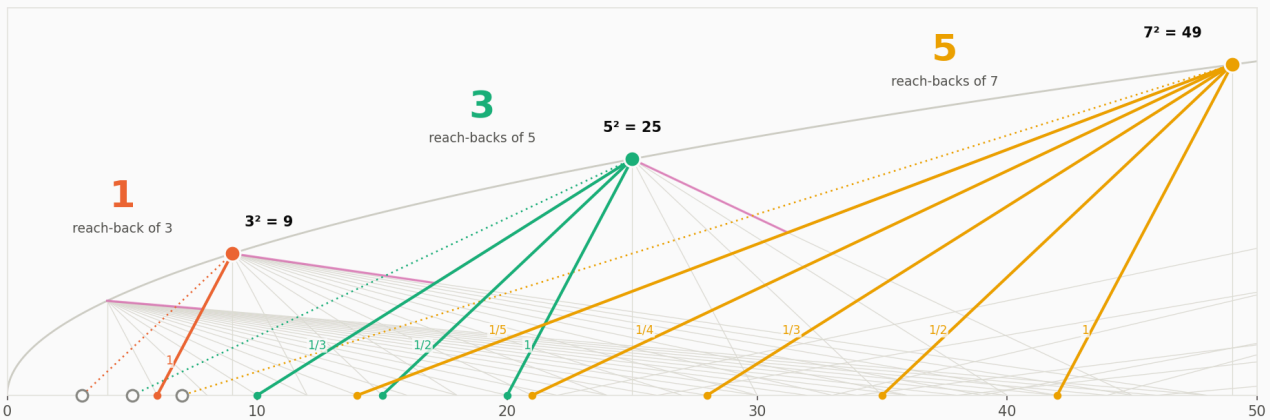
Part VI. Reach-backs: every rule

Every prime q 's rule is already running before its square. Its reach-backs are the $q - 2$ chords from (q^2, q) back to $2q, 3q, \dots, (q - 1)q$, with slopes $1/(q - 2), \dots, 1/2, 1$. The shallowest one is exactly the ramp of Part V. **Proven** (elementary).

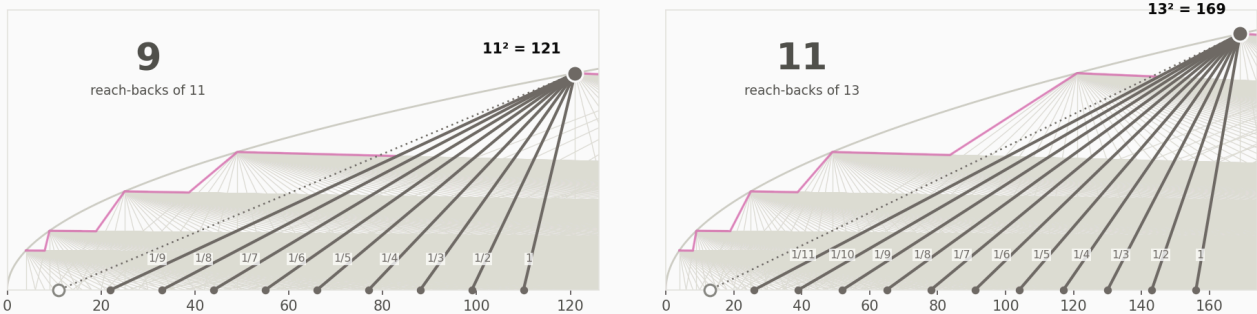
Reach-backs: every prime's rule is running before its square

Prime q 's reach-backs are its lines from (q^2, q) down to $2q, 3q, \dots, (q-1)q$. There are exactly $q-2$ of them: your 1, 3, 5. Their slopes are $1/(q-2), \dots, 1/2, 1$: every counting-number angle up to $q-2$. The shallowest one, to $2q$, is the ramp. The line to q itself (dotted) is never drawn, and that missing line is what keeps q prime. Every number a reach-back lands on was already crossed out by a smaller prime.

A. Your window. The counts you wrote are $q-2$. Magenta (faint) is the edge from the last picture: each ramp is the $1/(q-2)$ reach-back.



B, C. The next two primes. The count is 9, then 11, not 7: the reach-back count is $q-2$, so it only takes values two below a prime.



Reach-back counts: 1, 3, 5, 9, 11, 15, 17, 21, 27, 29, ... (every prime minus 2)

Sieve of Eratosthenes starts each prime at q^2 for exactly this reason: the $q-2$ reach-back hits are already done. They add no new records. They only set the angles.

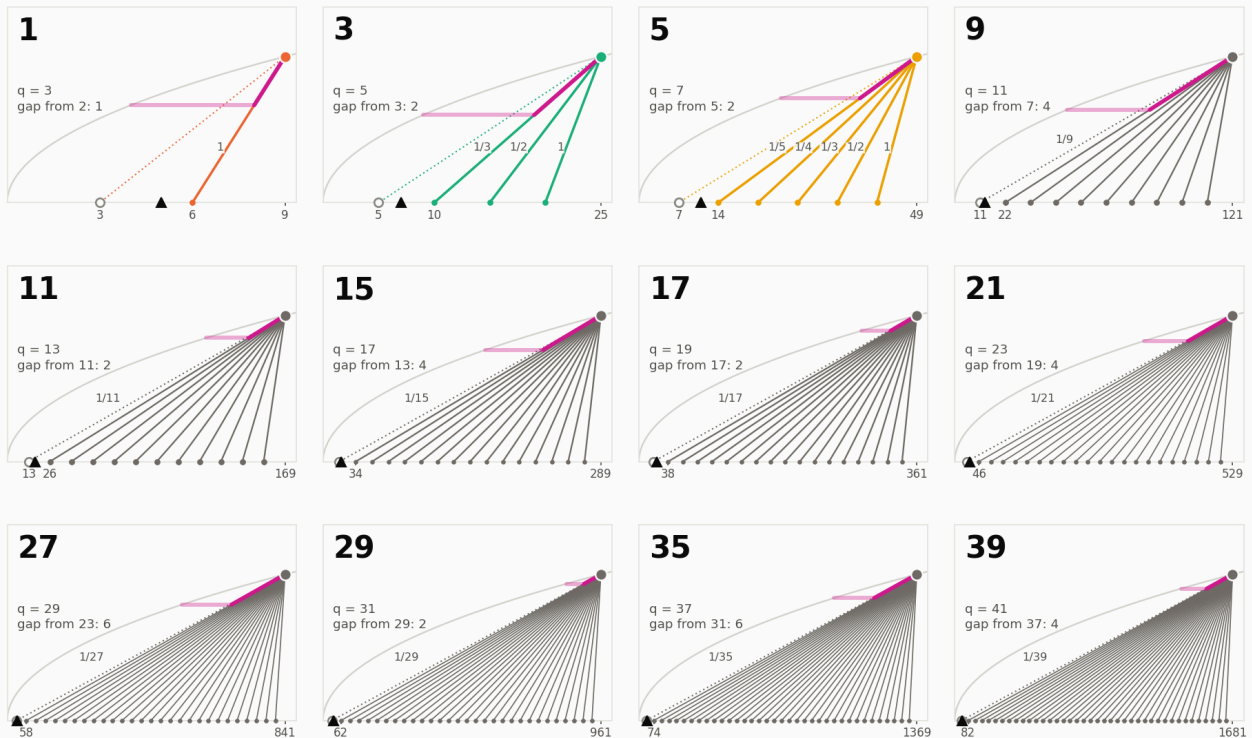
Figure 13. The counts Christophe wrote on the sieve (1, 3, 5) are the reach-backs of 3, 5 and 7, with slopes labeled. The dotted line from (q^2, q) to q itself is the one chord never drawn: that missing line is what keeps q prime. Every number a reach-back lands on was already crossed out by a smaller prime.

The timeline of one prime.

1. q appears on the axis at $x = q$, as a gap where no chord lands.
2. The next prime appears before $2q$ (Bertrand's postulate, proved by Chebyshev), so the next rule exists before q 's rule touches anything.
3. q 's first chord lands at $2q$; it makes $q-2$ reach-back hits, all on numbers already crossed out.
4. q 's first new record is q^2 , which is why the sieve of Eratosthenes starts each prime at its square.

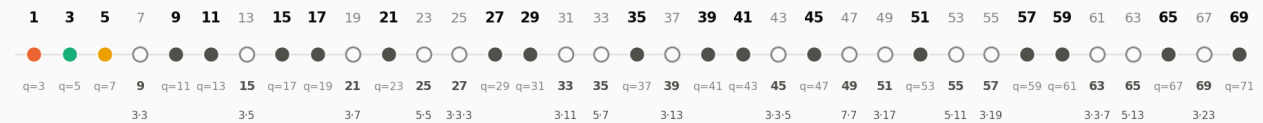
Every rule: the reach-backs of each prime

Each panel is one prime q in its own window, from 0 to q^2 . The big number is its count of reach-backs, $q - 2$. Magenta is the piece of the shallowest reach-back (the line to $2q$) that becomes the upper edge: the ramp from the previous prime's step up to (q^2, q) . Dotted: the line to q itself, never drawn. The small triangle on the axis is the next prime, which always shows up before $2q$, before q 's rule lands anywhere.



Why the count skips 7: an odd number k is a count only when $k + 2$ is prime.

Filled: a count ($k + 2$ is prime). Open: a hole, with the composite $k + 2$ written under it. 7 is missing because $9 = 3 \cdot 3$, which is rule 3's first new record.



Holes sit at 7, 13, 19, 23, 25, 31, 33, 37, 43, 47, ... : two below each odd composite. So the rule counts carry the primes; they do not generate them.

Figure 14. Every rule from 3 to 41, one panel each, with the count $q - 2$ in the corner, the magenta ramp, the dotted missing chord and the next prime (triangle). Bottom: the count ladder. An odd number k is a count only when $k + 2$ is prime; the holes (7, 13, 19, 23, 25, 31, 33, ...) sit two below the odd composites. The count 7 is missing because $9 = 3 \cdot 3$ is the rule of 3's first new record.

All the rules on one ruler

Horizontal axis is $\sqrt{x}$, so the parabola becomes the straight diagonal and every prime sits on it at (q, q) . Each fan is one prime's reach-backs, landing on $2q \dots (q-1)q$. Magenta is the upper edge: a flat step at each prime, then a ramp that is the next prime's shallowest reach-back. Straight lines in x bend on this ruler; nothing else changes. Numbers under the diagonal are the reach-back counts $q-2$.

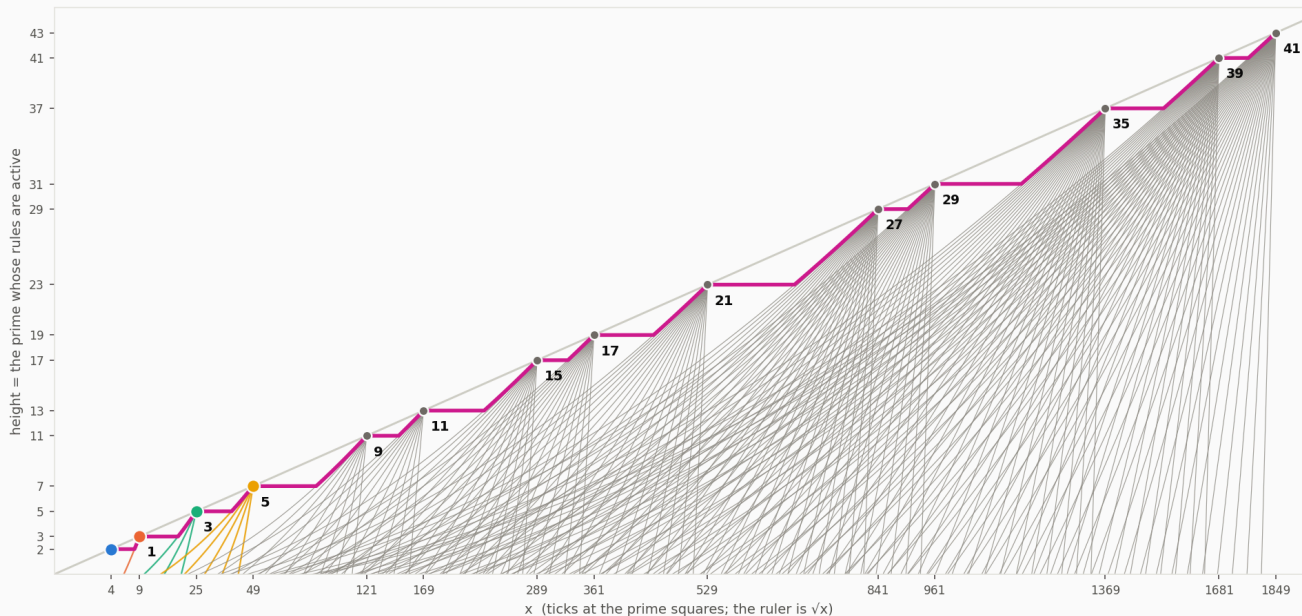


Figure 15. All the rules on a $\sqrt{x}$ ruler, where the parabola becomes the diagonal and every prime sits on it at (q, q) . The magenta staircase climbs the diagonal: a flat step at each prime, then a ramp that is the next prime's shallowest reach-back. Numbers under the diagonal are the counts $q-2$.

Honest reading. The reach-back counts inherit the primes rather than predict them: to draw a reach-back you must already know q is prime. The picture reads backward (the square in the future reaching into the past), but it is a receipt of q 's primality written backward, not a forecast. The magenta staircase contains every prime and every gap losslessly; drawing its next step without already knowing the next prime would amount to a formula for the primes. **Interpretation:** "primes existed before they existed" is exactly steps 1 to 4: presence precedes contribution.

Part VII. The magenta walk and the RH ceiling

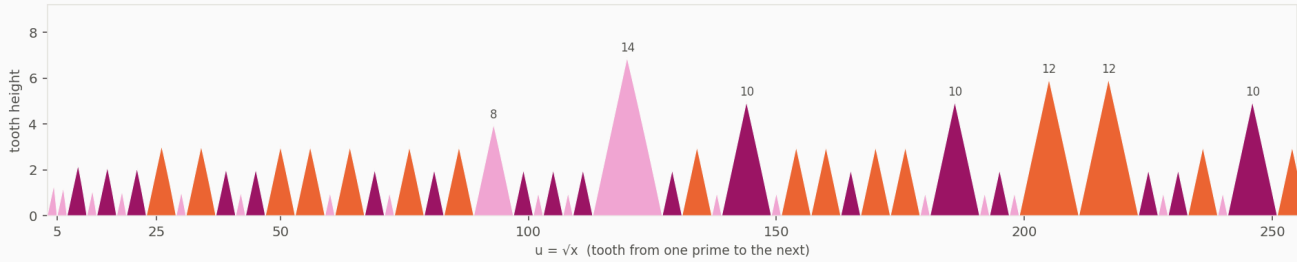
On the $\sqrt{x}$ ruler, each step of the magenta walk leaves one tooth between the diagonal and the staircase. Every tooth is the same triangle: up at 45° for half the prime gap, down at 45° for the other half, base g and height $g/2$. The walk is a chain of triangles whose sizes are the prime gaps. **Proven** in the limit (corrections of order $1/p$), **Computed** to 10^{11} .

The run. A segmented sieve written for this paper (C, two threads) reproduced $\pi(10^{11}) = 4,118,054,813$ exactly and all 40 published record gaps below 10^{11} , in 113 seconds. A 10^{12} run was started twice but did not finish: the cloud machine was reclaimed while idle.

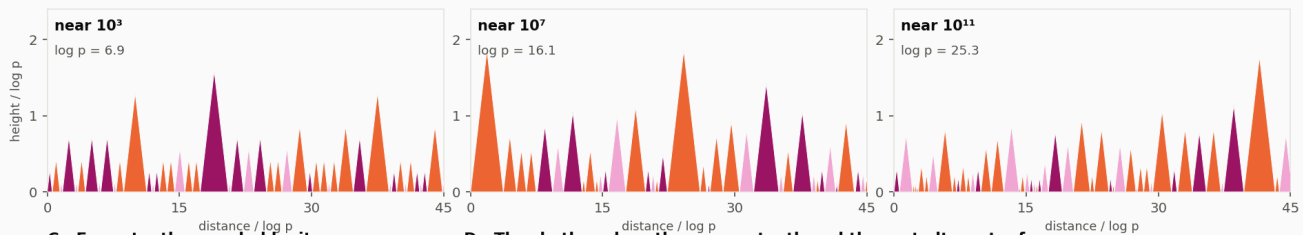
The magenta walk, mapped

On the $\sqrt{x}$ ruler each step of the walk leaves one tooth between the diagonal and magenta. Every tooth is the same shape: up at 45° for half the prime gap, down at 45° for the other half. So a tooth is a triangle with base g and height $g/2$, and the walk is a chain of triangles whose sizes are the prime gaps. Tooth size $= g/2$ is a whole number. Colors below: size a multiple of 3 (orange, the rule of 3), size $\equiv 1$ or $\equiv 2 \pmod 3$ (light and dark).

A. The walk from 3 to 257, tooth by tooth. Numbers are the gaps (8 and up); each tooth's height is half its gap.

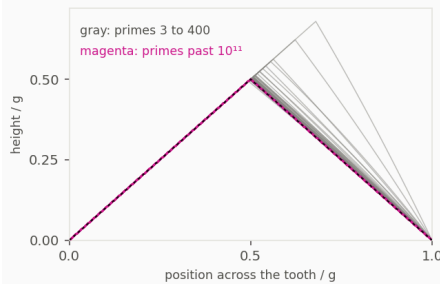


B. Same walk at 10^3 , 10^7 and 10^{11} , each window $45 \log p$ wide and scaled by $\log p$. The shape never changes; only the scale does.



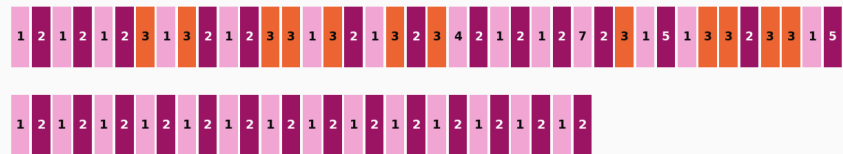
C. Every tooth, rescaled by its gap.

Small primes lean right (flat longer by $1 + 2/p$).
Past 10^{11} every tooth is the triangle.



D. The rhythm: drop the orange teeth and the rest alternate, forever.

Tooth sizes from $p = 5$ on. Row 2 keeps only non-multiples of 3 and prints the size mod 3.



Checked on every tooth up to 10^{11} : 4,118,054,810 teeth, 0 breaks. Steps from class 1 to class 5 (mod 6): 1,124,275,248; steps back from 5 to 1: 1,124,275,248. Every step out is matched by a step back. That is the rule of 3 acting on the walk.

E. The rules lift whole families of teeth. How often each tooth size appears near 10^{10} to 10^{11} , divided by a smooth baseline.

Black ticks: the prediction $\frac{1}{(r-1)(r-2)}$ over odd primes r dividing the size (Hardy-Littlewood). Size a multiple of 3: $\times 2$. Of 5: $\times 4/3$. Of 7: $\times 6/5$.

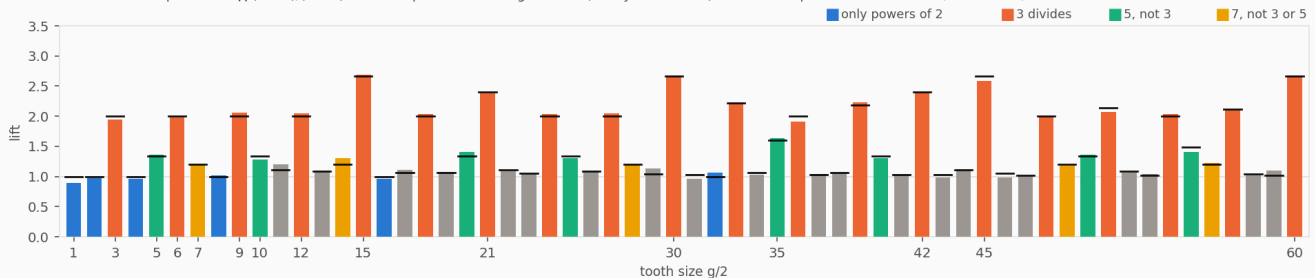


Figure 16. A: the walk from 3 to 257 tooth by tooth (numbers are gaps of 8 and up). B: the same walk near 10^3 , 10^7 and 10^{11} , scaled by $\log p$; the shape never changes, only the scale. C: every tooth rescaled by its gap collapses onto one triangle past 10^{11} (small primes lean right by a factor $1 + 2/p$). D: the rhythm. E: the rules lift whole families of teeth.

The rhythm (Christophe's 2, 4, 6). Set aside teeth whose size $g/2$ is a multiple of 3; the rest alternate 1, 2, 1, 2 (mod 3) forever. Checked on all 4,118,054,810 teeth past 5: **0 breaks**; steps from the class 1 (mod 6) to the class 5 number 1,124,275,248, exactly equal to the steps back. **Proven** (every prime past 3 is one below or one above a multiple of 6: a gap of 2 crosses from below to above, a gap of 4 crosses back, a gap of 6 stays on the same side). In gap terms: every spacing is 2, 4 or 6 plus extra 6s, and with the 6s removed the 2s and 4s take turns.

The rule lift. How often each tooth size appears near 10^{10} to 10^{11} , against a smooth baseline, matches the Hardy–Littlewood factor $\prod (r - 1)/(r - 2)$ over odd primes r dividing the gap: $\times 2$ for multiples of 3, $\times 4/3$ for 5, $\times 6/5$ for 7. Measured: size 3 at $\times 1.94$, size 15 (3 and 5) at $\times 2.68$ against 2.67, size 105 at $\times 3.20$ against 3.20; median deviation 2.2% over sizes up to 150. **Computed**; the factor is a **Conjecture** (Hardy–Littlewood) that fits the data closely.

Image — Figure 17. The walk against the RH ceiling.

Figure 17. A: tooth heights computed to 10^{11} (filled dots, tallest tooth per window) and published record gaps to 10^{20} (open dots), against the ceiling RH would impose. B: every record tooth on the Cramér scale $\log^2 p$. C: the energy of the walk, total tooth area up to X over $X \log X$.

Quantity	Value	Tag
RH ceiling on any gap after p	$(22/25) \cdot \sqrt{p} \cdot \log p$, so tooth $\leq$ half of that	Proven under RH (Carneiro, Milinovich, Soundararajan)
Room between the walk and the ceiling at 10^{20}	a factor of 2.2×10^8	Computed
Largest record gap / $\log^2 p$ past $p = 7$	0.9206 (gap 1132 after 1,693,182,318,746,371)	Computed from the published table
Record gaps stay under $\log^2 p$	all 85 published records past 7	Conjecture (Cramér scale)
Total tooth area / $(X \log X)$	0.261 at 10 , 0.337 at 10^4 , 0.407 at 10^8 , 0.429 at 10^{11}	Computed; limit $1/2$ is a Conjecture
RH bound on total tooth area	$C \cdot X \cdot \log^3 X$	Proven under RH (classical, Selberg)

Reading. The walk never approaches the RH ceiling; it hugs the Cramér scale. So RH caps the teeth, but proving they stay at the Cramér scale would be far stronger than RH.

Part VIII. The infinity plane and the quadratic plane

Both planes from the atlas are built correctly: every identity claimed for them holds on the real ξ at 16 to 40 digits. They give the cleanest place to state RH, and they do not by themselves supply its sign. **Computed** (identities), **Proven** (the RH equivalence, Sondow and Dumitrescu).

- **Infinity plane (atlas P02).** At one point z , record every derivative: $c_j = F^{(j)}(z)/j!$. The list decays faster than any power, and from one point it rebuilds F a full unit away to 40 digits.
- **Quadratic plane (atlas P03).** $G = c c^*$, every product of two derivatives. It is rank one, positive, and slides by $G(z + h) = T_h G(z) T_h^*$.
- **The sign entry.** $K = \operatorname{Re} G_{10} = \operatorname{Re}(F' \bar{F}) = \frac{1}{2} \partial_a |F|^2 = |F|^2 \partial_t \arg F$. RH is equivalent to $K \geq 0$ for every $a > 0$: $|\xi|$ never shrinks as you step right, away from the critical line ("no reversal").

Image — Figure 18. Checking the infinity and quadratic planes.

Figure 18. A: the derivative list at $z = 0.3 + 14i$ and its reconstruction error. B: receipts. C: outward growth $\Omega = K/|F|^2$ over 38,400 points (a from 0.002 to 1.5, t up to 60), smallest value 0.00009, positive everywhere. D: a zoom (t from 32 to 39) of the real ξ beside the same ξ multiplied by the atlas's scope polynomial Q (H9:17) planting one off-line zero at $a = 0.2$, $t = 35.5$: a teal reversal pocket appears.

Identity	Result
Translation law $c(z + h) = T_h c(z)$	agrees to 37 digits
Every 2×2 minor of G vanishes	to 5×10^{-16}
Rank one	one eigenvalue equal to the squared length of c , the rest about 10^{-22}
$G(z + h) = T_h G T_h^*$	to 2.5×10^{-16}
Rebuild c from one column of G	to 7×10^{-17} , up to phase
$K = \text{Re } G_{10}$, half the a -derivative of the squared size, and squared size times the t -derivative of the phase	all three equal $6.7857572987 \times 10^{-7}$

One labeling fix for the atlas. "Quadratic plane" names two different objects: P03 (the outer-product grid G) and, in the M18 entry, the square-coordinate map $w = z^2$ of P06. The atlas's own caution list already warns that "quadratic" has four meanings; renaming one of these two would prevent a transfer error.

Part IX. The fold, the teal pocket and the two mirrors

The faded band along the critical line is a literal fold. The functional equation makes the left half of the plane a mirror copy of the right, and the crease is where the two halves cancel: outward growth is exactly zero there. **Proven** ($\xi(s) = \xi(1 - s)$ and $\xi(\bar{s}) = \text{conj } \xi(s)$); mirror checked to 5×10^{-15} , **Computed**.

Image — Figure 19. The faded line is a fold.

Figure 19. A: real ξ on both sides of the line; the left half is the exact mirror of the right, growing away from the crease everywhere. B: the folded infinity plane: on the crease the quadratic grid becomes a perfect checkerboard (even derivatives purely real, odd ones purely imaginary), so $\text{Re } G_{10}$ is about 10^{-50} against a size of 10^{-9} ; 0.3 off the crease the checkerboard breaks and the sign entry turns on (4.4×10^{-10}). C: Davenport–Heilbronn, a function with the identical fold and no Euler product, has a zero off the crease at $0.8085 + 85.6993i$ with its mirror, and teal reversal pockets.

The fold alone cannot finish the job. It pins where the critical line is and forces any off-line zero to come with a mirror twin, but it cannot push zeros onto the crease. Davenport–Heilbronn proves this: same fold, zeros off the line. What ξ has and Davenport–Heilbronn lacks is the Euler product, the primes. **Proven.** Folding once more by $w = z^2$ (the atlas's P06) glues the halves and turns RH into "every zero of the folded function lies on one ray", the setting of Pólya's program and the de Bruijn–Newman constant (Rodgers and Tao: if RH holds, it holds only barely).

Image — Figure 20. What is inside the teal.

Figure 20. Horizontal slices of the landscape. Real ξ (right): every slice is a valley with its floor on the crease, and a zero is a hole at the valley floor. Davenport–Heilbronn (left): inside the teal pocket the crease flips from valley floor to ridge, with a sinkhole on each side (the off-line zero and its twin). The teal is the ground sloping down into those sinkholes.

Image — Figure 21. Mirrored across the x axis.

Figure 21. The second mirror. Every zero at height t has a twin at $-t$ (the bottom window is the exact reflection of the top one, checked). With both mirrors the picture centers on $s = 1/2$: on-line zeros come in pairs $\pm \gamma$, and an off-line zero comes in four, the corners of a rectangle centered on $1/2$.

Why fours matter. An off-line quartet adds to the total zero count without adding sign changes on the line, so it shows up as a shortfall. This is Turing's method, the way every zero up to height 3×10^{12} has been confirmed on the line (Platt and Trudgian). **Proven.** The mirrors fix the frame; they do not fix the zeros.

Part X. All zeros as one

Five exact ways to hold every zero at once, each built and checked on the real ξ . Each one restates RH precisely; none of them proves it. **Computed** (all five), with the RH equivalences **Proven** in the literature.

Image — Figure 22. Fold everything onto itself.

Figure 22. A: folding across both mirrors squeezes every off-line group of four into one point and every on-line pair onto one edge (the four Davenport–Heilbronn zeros land on the same point to 15 digits). B: all 1,000 first zeros of ξ sit on the edge; Davenport–Heilbronn’s point sits 52.9 off it. C: the zeros space themselves like the eigenvalues of random mirror-image (Hermitian) matrices: gaps below 0.2 occur 0.3% of the time for zeros, 0.8% for the matrices, 18.1% for random points (Montgomery, Odlyzko).

Image — Figure 23. All the zeros in one matrix.

Figure 23. A: a $2,000 \times 2,000$ mirror-image matrix holding the first 1,000 zeros and their twins, scrambled so every entry mixes all of them; its eigenvalues return the zeros to 4×10^{-12} . B: its characteristic curve rebuilds ξ on the line to 0.008% once zeros past the 1,000th are added as a smooth tail. C: the map $w = 1 - 1/\rho$ bends the critical line into the unit circle; all 2,000 values land on it (to 2×10^{-16}) within 4.05° of the single point $w = 1$. D: summing $1/(\rho(1 - \rho))$ over every zero gives $2 + \gamma - \log 4\pi = 0.0461914179\dots$, known without knowing one zero; 1,000 pairs plus a smooth tail land within 3×10^{-7} .

The matrix in Figure 23 is circular: it was built from known zeros. The Hilbert–Pólya prize is a mirror-image operator built from something else (the primes) whose values are the zeros. The circle picture carries Li’s criterion: raise every point to the n -th power and sum; the total stays positive for every n exactly when RH holds. Its catch is crowding: Davenport–Heilbronn’s off-line zero at height 85.7 misses the circle by only 4×10^{-5} .

Image — Figure 24. Every zero in one dot.

Figure 24. The one dot: the full derivative list of ξ at $s = 1/2$, a single point of the infinity plane with infinitely many coordinates. A: every odd coordinate is zero to 88 digits (the two folds); every even one is positive. B and C: reading the first N coordinates rebuilds ξ zero

by zero; 260 coordinates return all 14 zeros below height 62, each to six decimals. Low zeros live in the early coordinates, high zeros deeper in.

RH can be stated as a property of this dot alone: the Jensen polynomials built from its coordinates must have only real roots (Pólya; Griffin, Ono, Rolen and Zagier proved each fixed degree for large shifts). All 279 tested here (degrees 2 to 10, shifts 0 to 150) pass. The same test also passes for ξ with a planted off-line zero at height 35.5, so low degrees cannot see a zero that high.

Image — Figure 25. One plane, one dot.

Figure 25. Each zero divided into its own mirror image across the line. A zero on the line is its own mirror, so the ratio is exactly 1: one dot. Real ξ : all 1,000 zeros, each re-found from a start well off the line ($\beta = 0.62$), land within 10^{-31} of the dot. Davenport–Heilbronn, every zero up to height 200: 122 on the dot, 16 off it (four quartets, at heights 85.7, 114.2, 166.5 and 176.7), missing by 0.0072, 0.0026, 0.0009 and 0.0025.

Interpretation: Christophe's "one dot, one location, and anything off would not sit on it" is Figure 25 exactly. RH is the statement that the plane holds only that dot, forever.

Part XI. The horn of rings and where an off-line zero travels

Turn each zero's height into the circumference of a ring and stack the rings in order: the first 1,000 zeros make a horn that grows exactly by the zero-counting law. An off-line zero cannot become a ring, and the horn comes up exactly two rings short for each off-line pair.

Computed; the counting law is **Proven** (Riemann–von Mangoldt).

Image — Figure 26. Every zero as a ring.

Figure 26. A: the horn (every fourth ring drawn); the first ring has circumference 14.13, ring 1,000 has 1,419.4. B: ring radius against the exact counting law, and the growth per ring against the law $1/\log(\text{radius})$ (mean 0.2239 measured, 0.2244 predicted). C: rings counted minus the law. For ξ it wobbles inside ± 0.73 across 1,000 rings. For Davenport–Heilbronn it drops by exactly 2 at each off-line pair (85.7, 114.2, 166.5, 176.7), ending 8 short at height 200.

Interpretation, made precise. Christophe's "an off-line zero is a different object that does not exist as a zeta zero" has an exact form: it cannot be a ring. It remains a zero of the function and the counting law still counts it, so it shows up as a missing ring. Saying it cannot exist at all is RH itself.

Image — Figure 27. Where an off-line zero travels.

Figure 27. A: every zero is a wave through all the primes. The prime-count error divided by $\sqrt{x}$: the actual primes to 10^7 (gray), the first 1,000 on-line zeros out to 10^{12} (magenta), and the same plus one off-line group placed where Davenport–Heilbronn's sits (teal), which swells like a power of x : 1.9 at a million, 16 at a billion, 110 at a trillion. B: planting one off-line group changes every coordinate of the one dot: early coordinates by about 7% (height 35.5), 1.2% (85.7) and 0.1% (300); deep coordinates by more than 100%.

Does an off-line zero propagate backward?

Where	Backward?	What happens
The ring count	No	The shortfall starts at the zero's height and carries forward; the rings below are untouched (the atlas's H9:17 shows every on-line zero can stay in place).
The one dot	Yes, faintly	Every coordinate moves, the earliest ones by an amount that shrinks with the zero's height.
The primes	Yes, and louder with size	Its wave passes through every prime at every size and outgrows $\sqrt{x}$; on-line waves only creep by log factors.

The real primes show nothing like the teal swell, which is the classical statement that RH holds exactly when the prime error never outgrows $\sqrt{x}$ apart from log factors. **Proven.** So the primes are where an off-line zero would be loudest, which is why a proof has to lean on the prime side.

Part XII. The horn as a physical object, and the prime-side proof frontier

Treated as a real object, the horn has a measurable stiffness that the primes impose. The prime side has already proved a ladder of facts about it, and in August 2026 the share of zeros proved on the line jumped from 41.7% to 67.2% using Weil's Hermitian form, the same object this program is built on. RH itself still needs the energy positive for every probe size.

Image — Figure 28. The horn as a physical object.

Figure 28. A: measured stiffness. Number variance of the ring count in windows of L average spacings, from all 22,491 zeros up to height 20,000 (count checked against the exact law: 22,462 found above height 100 against 22,462.4 predicted). Random Hermitian matrices keep loosening (0.69 at $L = 30$); the zeros stop near 0.3 to 0.4. B: share of zeros proved exactly on the line, by year. C: how the two-thirds proof works. D: the program's route, the odd Weil energy floor against probe half-width a .

The stiffness is the primes. Past a few spacings the zeros stop loosening, while random matrices keep loosening logarithmically. Berry's semiclassical formula explains this: the primes act like the shortest closed paths of a vibrating system and lock the long-range order. **Computed** (measurement); Berry's explanation is a well-supported heuristic, tagged **Conjecture**.

What the prime side has proved about the object.

Fact about the horn	Status	Source
It grows by the counting law	Proven	Riemann–von Mangoldt
No zero near the outer walls: none with $\sigma \geq 1 - 1/(5.558691 \cdot \log t)$, $t \geq 2$	Proven	Mossinghoff, Trudgian and Yang
Every zero up to height 3×10^{12} is on the line	Proven by computation	Platt and Trudgian
At least 41.72% of all zeros are on the line	Proven, 2020	Pratt, Robles, Zaharescu and Zeindler

Fact about the horn	Status	Source
More than two thirds (67.2%) are simple and on the line	Proven, unconditional, August 2026; Lean-checked; read by Conrey and Goldston; wider review beginning	Alpöge and Furman, with an unreleased Claude model
Every zero is on the line (RH)	Open	Weil positivity for every probe

How the two-thirds proof works, in the horn’s language.

- 1. Compress Weil’s energy form into a finite matrix by probing one stretch of the horn (heights T to 2T) with a row of identical windows.
- 2. Each ring on the dot contributes one positive direction, worth 1 to the trace.
- 3. Each off-line pair contributes one positive and one negative direction; Sylvester’s law of inertia counts them. This is the teal pocket in matrix form.
- 4. The trace counts rings. The total energy (the matrix’s squared size) comes from the primes through sums of $\Lambda(n)^2$ known exactly (Montgomery), $4/3$ per ring for the simplest window.
- 5. A rank-trace inequality then forces at least $2 - 4/3 = 2/3$ of the rings onto the dot.

This is the same machinery as the program’s engine: finite Weil matrices and negative-eigenvalue counts. Anthropic states it does not expect these techniques to prove RH, and even 100% would not be RH, because a proportion still allows rare exceptions.

The program’s route (panel D). RH is equivalent to the odd Weil floor staying positive for every probe half-width a . The program’s CP22 bound covers $0 < a \leq 3/5$; Zhu’s is certified to 0.8; the numerics here reach 1.2. The floor falls super-exponentially: 1.94×10^{-4} at 0.5, 5.97×10^{-7} at 0.6, 1.57×10^{-14} at 0.8, 1.49×10^{-26} at 1.0, 4.76×10^{-45} at 1.2. A proof needs a prime-side mechanism uniform in a , not one support at a time.

Next steps that the machinery is already set up for.

- ☐ Rebuild the two-thirds argument inside the program’s Weil-matrix engine (same compression, trace and inertia count), as a receipt.
- ☐ Search for a better window: any window with a smaller prime-side constant than $4/3$ raises the proved share.
- ☐ Look for the uniform-in- a mechanism: the relay (Part I) shows each prime power is load-bearing at its entry, but not yet why the hand-off never fails.

Figure 30. A: the spectrum $F(t)^2$ of the best test function at four widths; dashed lines are the horizons, ticks are zeta zeros. B: distance from each zeta zero to the nearest zero of F , with height measured in horizons. C: the share of spectrum energy outside the horizon (blue) and the share of the floor paid by zeros inside it (orange). D: the floor against horizon height, a straight-line fit on $a \geq 0.8$, and Zhu's independent value at $a = 1.2$.

a	Horizon T^* (a)	Floor λ	Zero sum $\div$ λ	Floor share paid inside T^*	Energy share outside T^*	First zero matched to
0.4	14.0	1.47×10^{-2}	0.999993	0 (no zeros inside)	2.2×10^{-2}	not matched (0.3 off)
0.6	20.9	5.96×10^{-7}	0.999983	9.6×10^{-6}	7.1×10^{-6}	4.9×10^{-5}
0.8	31.1	1.56×10^{-14}	0.999995	1.0×10^{-7}	8.1×10^{-11}	2.8×10^{-12}
1.0	46.4	1.49×10^{-26}	0.999999	1.6×10^{-13}	3.5×10^{-18}	4.3×10^{-24}
1.1	56.7	7.68×10^{-35}	0.999983	8.2×10^{-16}	8.1×10^{-24}	2.6×10^{-32}

What the test shows.

- 1. The books balance.** The matrix floor equals the zero sum to within 2×10^{-5} at every width tested. That is Weil's explicit formula checked end to end on real data, and it confirms the engine's normalization. **Computed.**
- 2. The energy stays inside the cone.** All but 7×10^{-6} ($a = 0.6$) down to 8×10^{-24} ($a = 1.1$) of the spectrum's energy lies below the horizon. **Computed.**
- 3. The bill is paid outside.** Zeros inside the horizon pay at most 10^{-5} of the floor. The best test function parks a zero of its spectrum on every zeta zero inside the horizon (the first to 32 digits at $a = 1.1$), keeps tracking a little past it, and is off by 0.01 once the zeros reach 1.2 to 1.7 horizons. **Computed.** The tracking itself is known: Connes, Consani and Moscovici see it with the even eigenvector and primes up to 13 (first zero to 2.5×10^{-55}), and Groskin's May 2026 run of a related truncated form reaches about 300 digits for the first ten zeros.
- 4. The floor runs on the horizon's clock.** $-\log \lambda$ climbs about 1.83 per unit of T^* , so $\lambda(a) \approx \exp(-1.83 \times 2\pi e^{(2a)})$ up to a constant. Fitted on $a = 0.8$ to 1.1, it predicts about 1×10^{-44} at $a = 1.2$; Zhu's independent value is 4.76×10^{-45} . The rate per unit of T^* still creeps up (1.65 at $a = 0.6$, 1.91 at 1.08), so this is a fit, not a law. **Computed.**

Did the physics trick work? In physics, a finite speed plus an energy that never grows gives stability at every time. Both halves exist here, but they point the wrong way. The floor $\lambda(a)$ never grows as the cone opens (**Proven**: a width- a probe is also a probe of every larger width). Never growing is exactly what lets it sink toward zero, so on its own it gives no floor. The test also shows where a floor would have to come from: the primes inside the cone fix the form exactly, yet its sign is settled by zeros beyond the horizon. Read backwards, that is where an off-line zero would surface: once the horizon passes its height, the best test function can aim at it, and its term $2 \operatorname{Re} F(\gamma + i\beta)^2$ can be negative (the signature block of Part XII). **Interpretation**.

A sharp target. A prime-side proof that $\lambda(a) \geq \exp(-C \times 2\pi e^{(2a)})$ for one fixed C and every a would prove RH, since the floor could then never reach zero. Through $a = 1.2$ any $C \geq 1.5$ fits the data; the rising rate in finding 4 says how much room is left. **Open**.

- ☐ Run the same test on the Davenport–Heilbronn control. Its zeros are denser (conductor 5), so its horizon is $(2\pi/5) e^{(2a)}$, which reaches its off-line zero at height 85.7 near $a = 2.1$. Prediction, untested: its floor turns negative near there.
- ☐ Push the decay law past $a = 1.2$ with more modes and bits, to see whether the rate per unit of T^* settles.

Part XIV. Redshift, refraction and the constant across the expansion

Christophe asked whether cosmological redshift could be the primes showing through as light travels, what is expanding in the math, and how to hold a constant across it. What expands is the scale x itself. The prime waves redshift exactly the way light does in an expanding universe. In comoving units there is a constant, $2 + \gamma - \log 4\pi = 0.04619$ (γ is Euler's constant here), which the primes hit to within 1% out to 10^{10} and which survives only if every zero is on the line. **Proven** (the redshift law; the constant, given RH), **Computed** (the measurement), **Interpretation** (the physics reading).

What is expanding. The scale factor is x . Each zeta zero enters the prime count as a wave $\cos(\gamma \log x)$, and on the ordinary number line that wave's wavelength is $2\pi x/\gamma$. Between any two points every zero's wave stretches by the same factor x_2/x_1 , whatever its frequency. That is the signature of cosmological redshift: one factor $1 + z$ for every wavelength. In the Weil form the probe's reach $e^{(2a)}$ plays the scale factor, with the half-width a as cosmic time and expansion rate 2. **Proven** (one line of algebra).

The comoving recipe. Cosmology removes expansion with comoving coordinates. The prime version has three parts:

1. **Ruler.** Measure in $\log x$. The step dx/x is unchanged by stretching.
2. **Frequencies.** In $\log x$ each zero rings at exactly γ at every scale. The zeros are the quantities that never redshift.
3. **Brightness.** Divide by $\sqrt{x}$. Every on-line zero then has a wave of fixed size forever, while a zero at $\beta + i\gamma$ would still grow like $x^{(\beta - 1/2)}$. The $1/2$ is the unique rescaling that makes stretching energy-preserving, the same $1/2$ as in Berry and Keating's $(x^p + p x)/2$. RH is equivalent to $\psi(x) = x + O(\sqrt{x} \log^2 x)$ (von Koch, 1901). **Proven.**

Image — Figure 31. Taking the expansion out of the primes.

Figure 31. A: $\psi(x) - x$ on the ordinary number line up to 200,000. The first zero's wave is 15,558 long near 35,000 and 53,343 long near 120,000, and the height grows like $\sqrt{x}$. B: the same signal on the comoving ruler, divided by $\sqrt{x}$, from 10^2 to 10^{10} ; dotted lines mark the predicted spread ± 0.2149 . Orange adds one hypothetical off-line zero placed like Davenport-Heilbronn's ($\beta = 0.8085$, height 85.70). C: frequency content of the comoving signal in two windows about 1,800 times apart in x ; both peak within 0.04 of the first six zeta zeros. D: running average of the squared comoving signal against the constant $2 + \gamma - \log 4\pi$.

The constant across the expansion. If RH holds, the average of the squared comoving wave over $\log x$ converges to $\sum 1/|p|^2 = 2 + \gamma - \log 4\pi = 0.04619$ (Cramér, 1922; the same sum appears in Part X). With every prime power up to 10^{10} the running average is 0.0458, within 1% of the constant, and each decade from 10^2 to 10^{10} stays between 0.042 and 0.048. With one off-line zero added, the same average passes 35 by 10^{10} and keeps climbing. **Proven** (the limit, given RH), **Computed** (the measurement).

The Weil form's own comoving constant. In comoving units the horizon of Part XIII never moves: $T^*(a)/e^{(2a)} = 2\pi$. The floor gives a candidate constant: $-\log \lambda(a)$ divided by $T^*(a)$ is 1.39 at $a = 1.1$ and 1.47 at $a = 1.2$ (Zhu's value), still climbing toward the local rate of about 1.9. If it levels off, that level is the expansion constant of the energy floor. A proof that it stays below one fixed number for every a is the Part XIII target, and it proves RH. **Computed** (the values), **Open** (the bound).

Refraction, done right. Christophe's refinement: the energy is not lost but refracted, like light slowed in glass. Ordinary glass cannot redden light. It slows the wave and shortens its wavelength but never changes its frequency, so light leaves the color it entered. A

medium whose index keeps rising while the light is inside does change frequency ("time refraction"), and an expanding universe is mathematically that medium, with an index growing like the scale factor: the photon count is conserved and each photon's frequency drops. That is the precise sense in which "refracted, not lost" is right. **Proven** (standard physics).

As a mechanism for real starlight, absorbing and re-emitting fails on two observations. Distant images stay sharp, while re-emission would scramble directions. Redshift is the same for every spectral line to about a part in a million, while resonant absorbers bend colors differently and carve lines at their own frequencies. Supernova time dilation, the warmer cosmic background at high redshift, and the $(1+z)^4$ dimming of surface brightness already rule out any energy-loss-in-flight reading. These are measurements rather than proofs, and they all point the same way.

The zeros as absorbers. In the mathematics the absorbers are real:

- Connes' trace formula, built from the primes, shows the zeros on the line as an absorption spectrum; zeros off the line would enter as resonances instead.
- On the modular surface, a hyperbolic horn with one infinite cusp, waves scatter through $\xi(2s-1)/\xi(2s)$ (Faddeev and Pavlov; Lax and Phillips). Every zeta zero becomes a resonance: the wave is caught and released after a delay. A zero on the line at $\frac{1}{2} + i\gamma$ gives a resonance at $s = \frac{1}{4} + i\gamma/2$, so every one holds the wave for the same time. RH says every zero releases its light on one clock.
- Burnol, in the Lax-Phillips framework, restates RH (for all L-functions at once) as causality: an off-line zero would be a pole that lets the output precede its cause.
- Sierra (2014) built the physical version: a massless fermion seen by an accelerating observer, mirrors whose periods on the observer's clock are $\log p$, and the zeros as energy levels, reached in a limit with phases tuned from zeta itself.

Proven (the scattering identities), **Interpretation** (the reading as light).

- ☐ Push the comoving average to 10^{11} or 10^{12} with the same sieve and watch the 0.9% gap to 0.04619 close; the cross terms between zeros fade only like $1/\log x$.
- ☐ Compute the modular-surface time delay (the phase derivative of $\xi(2s-1)/\xi(2s)$ on its unitary line) and plot its peaks at half the zero heights.

Claim ledger

Every result in this paper, its tag and its receipt. Nothing here closes the open gate: positivity of the Weil form for every support.

#	Claim	Tag	Receipt
1	Each prime power from 2 to 9 is load-bearing: without it the odd Weil floor fails within 0.003 of its entry	Computed	420 to 640 digit arithmetic, 40 to 72 modes; floors match Zhu at 0.8, 0.9, 1.0
2	Color 2 pays 97.6% to 98.8% of the gamma plus pole deficit at $a = 0.8$ to 1.0	Computed	same engine
3	Record tunings close like $k^{(-1/d)}$ for d colors	Computed; law Proven (Dirichlet)	399 records for the octave and 3, to 209-digit k
4	Factorial minus all prime-power waves equals the zero bumps	Proven (explicit formula)	agreement 5×10^{-14} with 78,734 prime powers
5	The prime line envelope is $2x \cdot e^{(-x/2)}$, powers on $1/k$	Proven	algebra
6	Parabola chords land on products; primes are unhit	Proven	$x = ab + (a - b)y$
7	The sieve's upper edge is a staircase with flat $(q - p)(p + 2)$ and ramp $(q - p)(q - 2)$	Proven	matched to 0.072 on [4, 180]
8	A prime q has $q - 2$ reach-backs with slopes $1/1$ to $1/(q - 2)$	Proven	elementary
9	Tooth sizes (half-gaps) not divisible by 3 alternate 1, 2 mod 3	Proven	0 breaks in 4,118,054,810 teeth to 10^{11}
10	Tooth sizes are lifted by $\Pi(r - 1)/(r - 2)$	Conjecture (Hardy-Littlewood)	median deviation 2.2% near 10^{11}

#	Claim	Tag	Receipt
11	Record gaps stay under $\log^2 p$ past $p = 7$	Conjecture (Cramér scale)	all 85 published records; maximum 0.9206
12	Infinity and quadratic plane identities	Proven; checked	16 to 40 digits on real ξ
13	RH is equivalent to no reversal ($\text{Re } G_{10} \geq 0$ right of the line)	Proven (Sondow, Dumitrescu)	map positive on 38,400 points to $t = 60$
14	The fold alone cannot force zeros onto the line	Proven	Davenport–Heilbronn zeros at $0.8085 + 85.6993i$ and three more to height 200
15	An off-line quartet collapses to one point under the double fold	Proven	checked to 15 digits
16	Zero spacings match random Hermitian matrices	Conjecture (Montgomery pair correlation), strong data	1,000 zeros; gaps below 0.2 at 0.3%
17	The one dot at $s = 1/2$ holds every zero	Proven (entire function)	260 coordinates return 14 zeros to six decimals
18	Horn rings come up short by exactly 2 per off-line pair	Proven (counting law)	Davenport–Heilbronn: $-2, -4, -6, -8$
19	An off-line zero's prime wave outgrows $\sqrt{x}$	Proven	110 times $\sqrt{x}$ scale at 10^{12} in the model
20	Zero count stiffness saturates	Computed; Berry heuristic	22,491 zeros
21	More than two thirds of zeros are on the line	Proven, August 2026 (review ongoing)	Alpöge and Furman
22	A width- a probe feels only prime powers with $\log n \leq 2a$: a light cone	Proven (Paley–Wiener)	Figure 29

#	Claim	Tag	Receipt
23	The matrix floor equals the zero sum $2 \sum F(\gamma)^2$	Proven (Weil explicit formula); checked	within 2×10^{-5} over 22,491 zeros, $a = 0.4$ to 1.1
24	The best test function's spectrum stays inside $T^* = 2\pi e^{(2a)}$ and sits on the zeta zeros there	Computed; the tracking is known (Connes, Consani, Moscovici)	energy outside at most 7×10^{-6} ; first zero to 2.6×10^{-32} at $a = 1.1$
25	Zeros beyond the horizon pay the floor, which falls like $\exp(-1.83 \times 2\pi e^{(2a)})$	Computed (fit, not a law)	predicts about 1×10^{-44} at $a = 1.2$ against Zhu's 4.76×10^{-45}
26	Every zero's prime wave stretches by the same factor x_2/x_1 ; a redshift that is the same for every frequency	Proven	wavelength $2\pi x/\gamma$
27	On the comoving ruler the frequencies never redshift	Computed	two windows about 1,800 times apart peak within 0.04 of the first six zeros
28	The comoving brightness settles on $2 + \gamma - \log 4\pi = 0.04619$ (γ Euler's constant)	Proven given RH (Cramér); measured	0.0458 at 10^{10} ; with one off-line zero it passes 35
29	Zeta zeros are modular-surface resonances with one shared lifetime exactly when RH holds	Proven (Faddeev-Pavlov, Lax-Phillips)	resonance at $s = \frac{1}{4} + i\gamma/2$ for a zero at $\frac{1}{2} + i\gamma$
30	Every zero is on the line	Open	the program's gate

Methods and reproducibility

Every number above came from code run in these sessions; the table lists the engines, their precision and the check each one passed.

Engine	What it computes	Precision	Validation receipt
Odd Weil matrix, double precision	Q_a in 24 odd Legendre modes	16 digits	reproduces Zhu's floors at 0.5 to 0.8
Odd Weil matrix, high precision	exact rational kernel polynomials, inertia counts, inverse iteration for the lowest eigenvalue	420 digits (40 modes), 640 digits (72 modes)	floors at 0.8, 0.9, 1.0 match Zhu to 3 significant figures
Segmented prime sieve (C, two threads)	every gap to 10^{11} : histograms, records, residue transitions, tooth maxima	exact integers	$\pi(10^9)$ and $\pi(10^{11})$ exact; all 40 record gaps below 10^{11} match the published table
Riemann–Siegel root finder	all 22,491 zeros up to height 20,000	about 10^{-3}	count 22,462 above height 100 against 22,462.4 from the law
High-precision ξ (mpmath)	derivative lists by Cauchy transform, growth maps, the one dot	15 to 110 digits	$\xi(1/2) = 0.49712077818831\dots$; odd coordinates vanish to 88 digits
Davenport–Heilbronn	Hurwitz-zeta form with the standard constant κ ; line scan plus grid search and Newton	15 to 20 digits	zero at $0.80851718245 + 85.69934848538i$ reproduced; 130 zeros to height 200 against 129.5 smooth
Mirror-image matrix	$2,000 \times 2,000$ Hermitian from the first 1,000 zero pairs	16 digits	eigenvalues returned to 4×10^{-12}

Engine	What it computes	Precision	Validation receipt
Light-cone test	lowest-energy f at $a = 0.4$ to 1.1 ; its spectrum $F(t)$ by spherical Bessel sums below height 150 and the exact endpoint expansion above; evaluated on 22,491 zeros, the first 80 at 50 digits	640 bits	the two formulas for F agree to 10^{-99} at height 150; zero sum matches the floor to 2×10^{-5} ; at $a = 0.6$ the 72-mode floor (5.956×10^{-7}) sits 0.3% under the 40-mode one, as more modes can only lower a floor
Prime wave sampler (C)	$\psi(x)$ over every prime power at 32,840 log-spaced points up to 10^{10} (segmented sieve, compensated extended-precision sum)	exact integers; sums to 10^{-9}	matches a direct count to 8×10^{-9} at 2,018 points below 10^6 ; the 1,000-zero explicit formula tracks the data with correlation 0.983; 10^{10} in 61 seconds

Limits. The relay and all Weil floors beyond the certified ranges are numerics, not certificates. The 10^{12} sieve run did not complete. Zero statistics use Riemann–Siegel accuracy (about 10^{-3}), ample for spacing work but not for certification. Berry's saturation, Montgomery's pair correlation and the Hardy–Littlewood lift are tested, not proved. The color palette follows the program's convention: 2 blue, 3 orange, 5 green, 7 yellow, magenta for the walk, teal for reversal.

Sources

Pages opened during these sessions:

- [More than two thirds of the zeros of the Riemann zeta function are simple and on the critical line](#), Alpöge and Furman, August 2026
- [Claude has improved on a longstanding lower bound for the fraction of zeros of the Riemann zeta function that satisfy the Riemann hypothesis](#), Anthropic, August 2026
- [Zeta Spectral Triples](#), Connes, Consani and Moscovici, November 2025
- [High-Precision Approximation of Riemann Zeros via the Truncated Weil Form](#), Groskin, May 2026

- [The Riemann zeros as energy levels of a Dirac fermion in a potential built from the prime numbers in Rindler spacetime](#), Sierra, 2014
- [Scattering and number theory](#), Watkins (Faddeev–Pavlov, Lax–Phillips, Burnol)
- [Connes' trace formula and the Riemann zeros](#), Watkins
- [Fourier optimization and prime gaps](#), Carneiro, Millinovich and Soundararajan
- [Explicit zero-free regions for the Riemann zeta-function](#), Mossinghoff, Trudgian and Yang
- [The Riemann hypothesis is true up to \$3 \cdot 10^{12}\$](#) , Platt and Trudgian
- [More than five-twelfths of the zeros of \$\zeta\$ are on the critical line](#), Pratt, Robles, Zaharescu and Zeindler
- [A monotonicity property of Riemann's xi function and a reformulation of the Riemann Hypothesis](#), Sondow and Dumitrescu
- [Horizontal Monotonicity of the Modulus of the Riemann Zeta Function and Related Functions](#), Matiyasevich, Saidak and Zvengrowski
- [The De Bruijn–Newman constant is non-negative](#), Rodgers and Tao
- [Zeta Zeros on the Critical Line](#), Goldston and Suriajaya
- [On the difference between consecutive primes](#) (RH bound on the sum of squared gaps)
- [Prime gap: table of maximal gaps](#), Wikipedia

Further classical references named in the text (not opened in these sessions): Weil's explicit formula and positivity criterion (1952); Riemann–von Mangoldt counting law; Montgomery's pair correlation (1973); Berry, Semiclassical formula for the number variance of the Riemann zeros (1988); Hardy and Littlewood's prime-pair conjecture; Cramér's conditional gap bound; Baker, Harman and Pintz on prime gaps; Li's criterion; Pólya and Jensen polynomials (Griffin, Ono, Rolin and Zagier); Matiyasevich and Stechkin's parabola sieve; Zhu's certified odd Weil floors (arXiv 2608.24827, used in the first session); von Koch (1901) on the prime error term; Cramér (1922) on its mean square; Berry and Keating's xp Hamiltonian.

The program's own record: *Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas* (through Checkpoint 20), with Checkpoints 21 and 22.